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Traffic Sign Detection for Autonomous Vehicles

By

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Tan Pei Shi

Report is kind of too long, verbose, ..
50 to 60 pages are desired.

A PROPOSAL

SUBMITTED TO

Universiti Tunku Abdul Rahman

in partial fulfillment of the requirements

for the degree of

BACHELOR OF COMPUTER SCIENCE (HONOURS)

Faculty of Information and Communication Technology

(Kampar Campus)

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ABSTRACT

Done by: Ang Jia Pei The abstract should be short. Half a page should be enough.

This project aims to develop a reliable traffic sign detection system for autonomous vehicles in order to reduce road accidents and improve road safety. The system utilizes advanced image processing techniques, such as color identification and shape segmentation, to accurately detect and recognize traffic signs under various challenging scenarios, for example, complex backgrounds, various lightning Typo? conditions, damaged traffic signs, blurred traffic signs, and color fading in the traffic report Why it is new? signs. To address such limitations, in this ~~paper~~, we propose a new traffic sign detection system by employing techniques like HSI color space conversion, Gaussian filter, and region of interest (ROI) extraction to improve the accuracy of detection and precise classification of traffic signs.

The system captured inputted images with a camera in the vehicle, which are then preprocessed. Then, these images originally in RGB color space format are converted to HSI color space, later categorized based on hue, saturation, and intensity, allowing for color distinction. To further improve the images' quality, a Gaussian filter is then applied to reduce noise. The images proceed to the template matching stage. This system further identifies the traffic signs in images by using the predefined templates in order to isolate possible regions of interest. Then, it extracts relevant features such as edges, corners, and textures from these regions to perform analysis. Besides, this project applies the multiscale attention to ensure both large and small signs are accurately identified. Finally, ROI extraction is adopted to isolate the relevant regions for precise identification, and shape classification is implemented to identify and categorize the shapes of traffic signs that have been detected. The objective of the project is to enhance the efficiency of Advanced Driver Assistance Systems (ADAS)

by providing real-time detection and accurate traffic sign information, thus reducing human error and promoting efficiency.

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LIST OF ABBREVIATIONS

CNN	Convolutional Neural Network
GTSRB	German Traffic Sign Recognition Benchmark
NCC	Normalised Cross-Correlation
ROI	Regions of Interest
RGB	Red, Green, Blue
RPN	Region Proposal Network
HSI	Hue, Saturation, Intensity
HOG	Histogram of Oriented Gradients
LSS	Local Self-Similarity
LBP	Local Binary Patterns
SVM	Support Vector Machine
EDCircles	Enhanced Detection of Circles
TSDR	Traffic Sign Detection and Recognition

CHAPTER 1 – Introduction

1.1 Problem Statement and Motivation

Done by: Cheng Puei Ying

In recent years, the **detection and recognition of traffic signs** have become crucial components in the development of autonomous vehicles and advanced driver assistance systems (ADAS). These systems rely on accurate and reliable traffic sign detection to ensure the safety and efficiency of autonomous driving by providing real-time information about traffic regulations and warnings. However, this task is challenging due to various factors such as **varying lighting conditions**, **occlusions**, **complex backgrounds**, and the presence of **multiple signs in a single scene**. Traditional methods often struggle with these challenges, leading to inaccurate or missed detections. For instance, **changes in illumination** drastically affect the visibility and color perception of traffic signs, while **physical obstructions** like trees, buildings, or other vehicles can partially or fully block signs from the view of cameras. Additionally, the coexistence of multiple signs within a small area can confuse detection systems, making it difficult to identify and prioritize the most relevant information.

Therefore, this proposal aims to **develop a robust system that can accurately detect and recognize traffic signs under varying and challenging conditions** using advanced image processing and machine learning techniques. Such systems are vital for the seamless integration of autonomous vehicles into everyday traffic, reducing human error and enhancing overall traffic management. The successful implementation of this project will help **improve roadway safety and** the **reliability of autonomous vehicles**, facilitate the seamless integration of autonomous vehicles into everyday traffic, and **reduce human error**. Ultimately, the development of a robust traffic sign detection system is a technological pursuit of great societal importance. It promises to transform the way drivers navigate and interact with the world, improving everyone's quality of life by making roads safer and more efficient. This research is a critical step towards a future where technology and transportation will seamlessly integrate to improve the safety and efficiency of roads.

1.2 Project Scope

Done by: Tan Pei Shi

The description here is a bit short of setting the boundary of what is in and out of scope.

This project develops a traffic signs detection system for autonomous vehicles, which is capable of performing effective traffic sign recognition based on prior sign detection in terms of colour and shape. The system is designed to deliver a real-time output image with the type and position of each sign clearly labelled. In order to train and evaluate the performance of the proposed system, a standardised dataset, namely the German Traffic Sign Recognition Benchmark (GTSRB) and an additional 1000 real-world images will be utilised. It is assumed that each traffic sign captured from real life will have a standardised appearance in terms of shape, colour and symbol. Therefore, the colours to be detected for the additional images are limited to red, blue and yellow since most traffic signs are designed in these colours. However, there will be no fixed number of signs appearing in the same image, which means that some images may contain more than one traffic sign to be recognised simultaneously. Additionally, some images may include signs that are partially occluded or exposed to adverse weather or lighting conditions, in order to demonstrate the effectiveness of the feature-based image processing techniques proposed in the project.

1.3 Project Objectives

Done by: Tan Pei Shi

The project aims to develop a robust traffic sign detection system that can accurately identify and recognise various traffic signs under different conditions. Since the focus of the project is more on solving the difficulties of detecting and recognising partially obscured traffic signs and performing detection under bad weather, such as heavy rain, traffic signs that are more than 50% obscured are not considered in the project.

In order to evaluate the performance of the system, the GTSRB dataset will be used due to its complexity, which provides a challenging benchmark for detection, such as:

- Illumination changes: The GTSRB dataset provides various images with different light intensities, making it difficult for the system to perform detection.
- Image quality: Some captured images may be blurred or of poor quality due to the speedy movement of vehicles in real life.
- Occlusion: Traffic signs in some images may be occluded by other objects, such as trees, lorries and even occluded by other traffic signs.

Although the GTSRB is a valuable evaluation dataset, it still has some potential limitations, such as the lack of certain local traffic sign. To address this issue, an additional 1000 images have been included in the evaluation process. The project objective can therefore be broken down into several more specific objectives:

- The system should be capable of detecting and recognising all traffic signs in an image within a time frame of two seconds in real-time.
- The system should achieve an accuracy rate of 97% in recognising traffic signs under normal conditions (e.g., not being partially obscured by other substances).

- The system should achieve an accuracy rate of 90% for traffic sign recognition under unfavourable conditions, such as heavy rain, partial occlusion, and excessive or insufficient exposure to light.

1.4 Impact, Significance and Contribution

Done by: Ang Jia Pei

What is "it"? You mean the rate from previous statement?

Nowadays, the accidents and fatalities rate are steadily increasing. Although it is a leading cause of accidents, many people overlook it. Therefore, the development of autonomous vehicles and advanced driver assistance systems (ADAS) with traffic sign detection features ensures that drivers on the road are provided with instructions from traffic signs, such as when to slow down their vehicle. This project applies the **color identification and shape segmentation** techniques that help to **enhance the overall accuracy of traffic sign recognition**. Besides, this project improves the traffic sign recognition and segmentation by **overcoming limitations in pass work**, such as being able to detect and recognize the traffic signs in **various conditions** such as **complex background, blurred traffic signs, damaged traffic signs, color fading** of the traffic signs, **multiple signs** in a single scenario. These enhancements improve the overall performance of Advanced Driver Assistance Systems (ADAS), which play an essential role in **lowering the traffic-related death rate**. In addition, this project aims to help drivers have a better understanding of the instructions provided by different traffic signs and **reduce human errors.**

Moreover, the traffic sign detection system also **benefits visually impaired individuals**; for example, this system is able to provide accurate information about the surroundings to those visually impaired individuals, ensuring their **safety on the road**.

This project not only improves safety measures in autonomous cars, but it is also able to help those visually impaired individuals. To conclude, this project plays a vital role in promoting the **safety of transportation** systems and offering valuable contributions to the ongoing evolution of **automotive technology**.

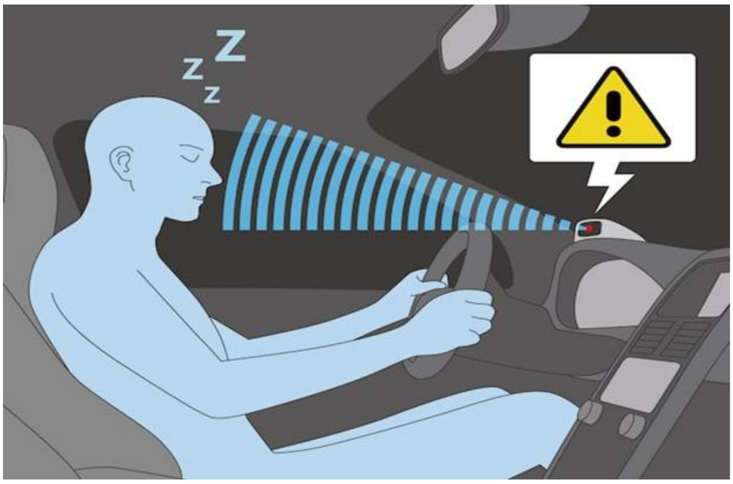

Figure 1.4.1 Driver fatigue necessitates a traffic sign detection system to alert the driver

Each figure should be cited and being explained in text.

Figure 1.4.2 Traffic sign detection alerts driver to stop

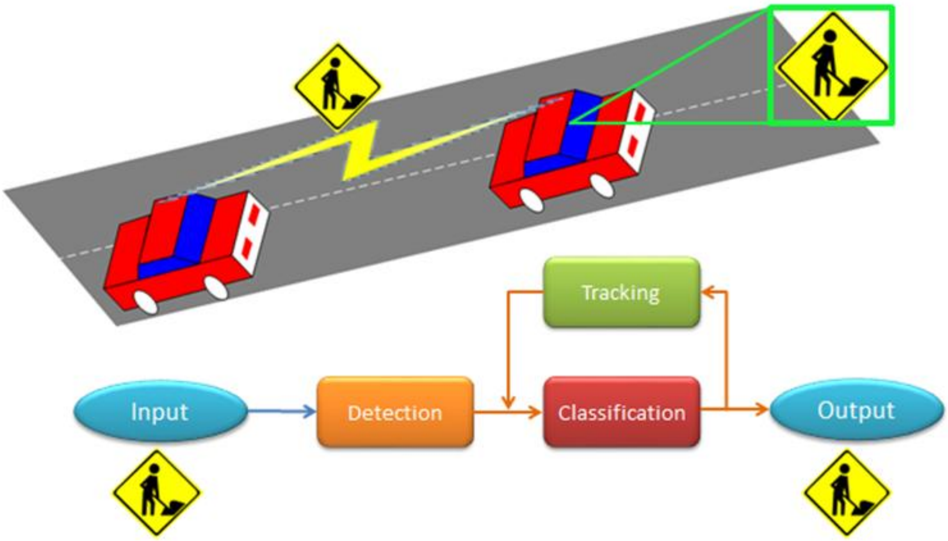

Figure 1.4.3 Flowchart for the traffic sign detection and recognition process

1.5 Background Information

Done by: Kok Yung Thow

Traffic sign detection and recognition (TSDR) is system that detect and recognize the traffic signs to assist the driver in various condition. **Traffic signs** are road infrastructures placed along the roadways to serve as visual signals to deliver messages to road users. It plays an important role in maintaining order and safety for the road users. Traffic signs are categorized into different types, such as guide signs, warning signs, and regulatory signs, based on the symbols, shapes, and colours to transfer the information effectively. The traffic signs might be different in terms of colours and shapes, depending on the countries, but the functionalities are the same. Figure 1.4.1 shows the examples of the traffic signs found in United States.

Where to find?

Figure 1.5.1 American Signs Examples in [5]

Traffic Sign Detection (TSD) serves as the first process of TSDR system that detecting the road signs from images or video frames recorded from the cameras integrated in vehicles. This process acts as a core fundamental factor to determine the success of autonomous driving as the detection accuracy will influence the traffic signs recognition process. It is generally implemented through computer vision and machine learning techniques. TSDR systems has been implemented using two major technologies today, including computer vision and machine learning techniques.

Computer vision is a field of study in data interpretation, analysis, and visualization of computers from reality. Its subsets, which are **image processing** and **object detection,** are applied in TSDR to enhance the visibility of traffic signs and to identify them in video frames effectively.

For **machine learning**, it is Artificial Intelligence (AI) subfield that focuses on developing models that learn patterns from datasets to enable computers to make predictions or decisions. Techniques such as **feature extraction**, **classification,** and

dataset training allow the identification of traffic sign features, classify them into different categories, and increase the accuracy and efficiency using datasets.

Image processing techniques with colour-based approach, and shape-based approach methods have been widely used in detecting the traffic signs in TSDR. Since either of them are rely on the high image quality to be well performed [5], the demand on TSD systems that can handle well with real-world conditions such as lighting, weather, and occlusions presence that causing poor traffic sign image quality, is growing due to the trend of autonomous driving and AI revolution.

In recent years, many TSDR studies have started applying the **hybrid mode of colour-based and shape-based techniques** in the first procedure in TSDR, as well as implementing **deep learning** techniques in many TSDR systems [5]. Deep learning refers to a subset of machine learning that uses computer algorithms to directly learn the characteristics and patterns of the traffic sign images from datasets, leading to better performance in changing environments of the real world.

CHAPTER 2 - Literature Reviews

2.1 Review on Colour Processing

Done by: Ang Jia Pei

Colour identification is a key component that helps with traffic sign recognition. However, developing effective traffic sign detection and recognition methods faces **several challenges, such as adverse weather conditions, noise from natural surroundings, colour fading, and shadows obscuring the signs**. According to [1] and Figure 2.1.1, **colour invariants and pyramid histogram of oriented gradients (PHOG) features** had been introduced.

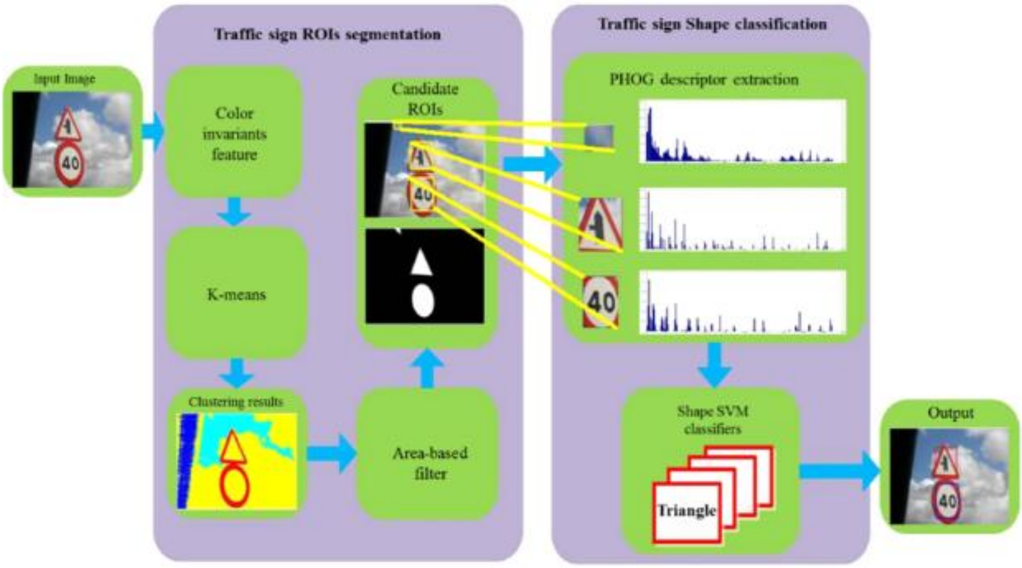

Figure 2.1.1 Block diagram of the proposed method in [1]

For colour invariants that proposed in [1], it can improve the recognition results under different lightning conditions or weather. By utilising the Gaussian colour model that was translated from the RGB colour model, the system can robustly segment images to identify candidate regions of interest (ROIs) and isolate traffic signs from complex backgrounds. The formula below shows the linear translation from RGB to Gaussian color model:

$$\begin{bmatrix} E \\ E_{\lambda} \\ E_{\lambda\lambda} \end{bmatrix} = \begin{pmatrix} 0.06 & 0.63 & 0.27 \\ 0.3 & 0.04 & -0.35 \\ 0.34 & -0.6 & 0.17 \end{pmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} \tag{1}$$

The E, E_{λ}, $E_{\lambda\lambda}$ denote the intensity, blue–yellow and green–red channel. To handle various imaging conditions for example viewing direction, surface orientation, illumination intensity and direction, formula such as $C_{\lambda} = E_{\lambda} / E$ and $C_{\lambda\lambda} = E_{\lambda\lambda} / E$ are applied for image color representations. The **K-means algorithm** is used to cluster the color invariant feature set f from an input image, then, segment it into different regions. **Hill-climbing algorithm** which act as an 8-neighbor search window to determine the K-seeds (largest bin within that window, local maximum values) in a two-dimensional histogram. Then, the image divided into different regions using a connected component algorithm on these pixels of each cluster spatially. However, colour invariant methods sometimes may **distort the perceived colours of an object**, which may potentially affect the accuracy of results in traffic sign recognition. For the PHOG, Canny edge detector is used to obtain edge contours, which are then divided into cells at different pyramid levels, and all the HOG vectors concatenated and create the final PHOG descriptor. Hence, Chromatic-Edge, which was introduced in [1], is a more advanced version of the Pyramid Histogram of Oriented Gradients. The **Otsu method** is applied to filter out weak edges and reducing noises. The improved PHOG descriptor is obtained by concatenating all HOG vectors at each pyramid resolution. The result was shown in Figure 2.1.2.

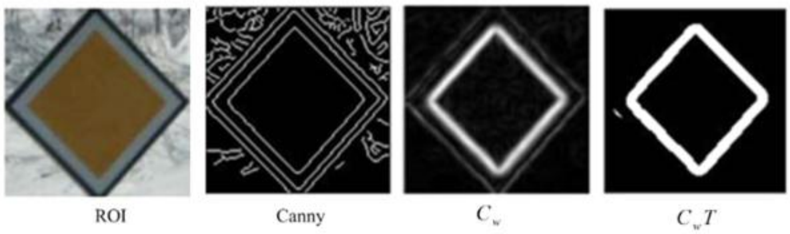

Figure 2.1.2 PHOG for a diamond shape traffic sign [1]

But, in **complex scenes with overlapping colours or textures, there may be a be a decrease in the rate of accuracy in edge detection**. Furthermore, when PHOG encounters a blurred or rotated traffic sign, it performs poorly.

According to [2], Figure 2.1.3 shows that HOG algorithms had been deployed by incorporating various colour spaces and extending the feature vector to include colour information. By incorporating colour information, the overall performance of HOG algorithms had been improved and got a better result when correlated with CIELab and d YCbCr colour spaces. This could be possibly due to the availability of two specific colour channels that align with the traffic signs. The study finds that **HSV and RGB spaces are less ideal for this application since RGB are very sensitive to environmental factors** such as lighting. For better interpretation of gradient and colour features according to their spatial context, the proposed system had deployed the ability to incorporate contextual information from surrounding pixels. On the other hand, to handle the large variation in background appearance, they try to reduce the memory usage while increasing the background information by applying the novel iterative Support Vector Machine (SVM) training paradigm, further improving the detection accuracy. Furthermore, the HOG-based method generally outperforms a state-of-the-art specific detection algorithm in many scenarios. Unfortunately, the **dimensionality of the feature vector and the computational complexity of the detection process**

had increased due to the incorporation of colour information. Hence, when the processing speed is critical, then it could be a limitation for real-time applications.

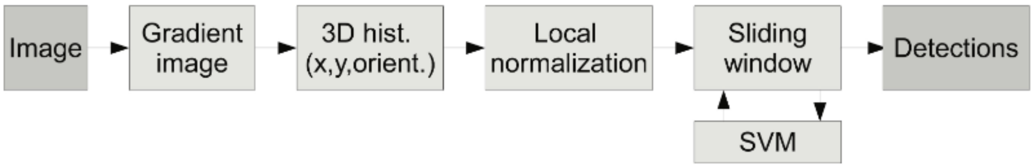

Figure 2.1.3 Flowchart for HOG algorithm [2]

In conclusion, [1] proposed the implementation of colour invariants and the Pyramid Histogram of Oriented Gradients (PHOG) to enhance recognition in complex scenes, such as varying lightning conditions and weather. **PHOG is an extension for the HOG descriptor**, by adding a **spatial pyramid structure by dividing the image, then performing concatenation**. Although they improved the results of recognition, these approaches pose some limitations, such as **perceived colour distortion and diminished accuracy in edge detection**. On the other hand, [2] suggests an alternative approach that employs HOG algorithms. These algorithms are more effective in the sense of being simpler and computationally efficient because they integrate colour information and concurrently increase the number of feature vectors.

2.2 Review on Shape Detection

Done by: Cheng Puei Ying

In addition to the detection of vibrant colors of road signs, one can also use the **shape of road signs** to carry out detection. The common shapes of the road signs that can be used for detection are circle, triangle, rectangle, and octagon. Instead of using conventional machine learning algorithms, the authors of [3] used an innovative shape detection method that was different from traditional techniques, which is **invariant geometric moments**, to perform shape classification.

According to [3], these moments were mathematical descriptors that **captured essential shape features** while being **independent of changes in scale, rotation, and translation**. This invariance was particularly beneficial for traffic signs, which could appear in different sizes and orientations depending on the viewing angle and distance. This technique simplified the detection process by **directly matching segmented ROIs to these known shapes**, thus avoiding the large amounts of training data and complex model tuning associated with machine learning approaches. In addition, geometric moments provided a compact and efficient representation of shape information, which sped up processing and reduced computational requirements.

The proposed system by [3] consisted of three basic steps. The first step involved **image segmentation** using thresholding of components in the HSI color space to extract ROIs. This technique exploited the distinctive color information of the traffic sign to separate it from the background and was particularly effective due to its robustness to different lighting conditions. Next, the system focused on **detecting the geometric shapes** of the segmented ROIs, as traffic signs were usually limited to shapes such as triangles, circles, rectangles, and octagons. This approach utilized invariant geometric moments to match ROIs with standard traffic sign shapes.

	Red	Blue
Hue	$0 \leq H \leq 10$ Or $300 \leq H \leq 360$	$190 \leq H \leq 260$
Saturation	$25 \leq S \leq 250$	$70 \leq S \leq 250$
Intensity	$30 \leq I \leq 200$	$56 \leq I \leq 128$

Figure 2.2.1 Threshold values used to extract ROIs with blue and red colors.

All figures should be cited and explained in text.

$$I(A, B) = \sum_{i=1}^{7} |m_i^A - m_i^B|$$

where m_i^A and m_i^B are defined as follows:

$$m_i^A = sign(h_i^A) \log |h_i^A|$$

$$m_i^B = sign(h_i^B) \log |h_i^B|$$

Figure 2.2.2 Metric used to match the ROIs with the appropriate shape classes.

The final step of the system was to identify specific traffic signs **based on the pictograms and symbols** appeared on them. To address this, [3] introduced a new descriptor that combined several feature extraction techniques, including HOG extended to HSI color space, LSS features, and LBP. The integration of these features enabled the descriptor to efficiently utilize the **color, texture, and structural information** that was critical for distinguishing between traffic signs that were **similar in shape but different in color**, such as warning and prohibition signs.

In [3], the **HOG feature** utilized the gradient information in the HSI color space to **capture directional intensity variations**, while the LSS feature provided insight into the **structural layout and repetition patterns** of the sign. The addition of LBP further enhanced the descriptor by **capturing local texture information**, which was valuable for identifying subtle details in traffic sign pictograms. This multifaceted approach ensured that the descriptors were robust to color, shape, and texture variations, thereby improving the accuracy of the recognition process.

The combined features were then classified using **Random Forest algorithm**, which was chosen for its unique strength in handling complex feature sets with high-dimensional data. Random Forests, an ensemble method consisting of multiple decision trees, provided additional robustness and generalization capabilities. It enhanced classification accuracy by **aggregating predictions from various trees**, each constructed from different bootstrap samples and subsets of predictors. This ensemble approach mitigated overfitting and performed well in the presence of noise and variability.

In short, [3] presented a robust method for TSDR that integrated image segmentation, shape detection, and sign recognition using advanced techniques to offer an accurate and efficient recognition, which was essential for autonomous vehicles and advanced driver assistance systems. The system employed the **HSI color space for robust image segmentation** and used **invariant geometric moments for shape detection**. The recognition phase combined HOG features, LSS features, and LBP to form a comprehensive descriptor. This multifaceted approach enhanced the system's robustness and generalization capabilities. However, the system's reliance on multiple feature extraction techniques resulted in **high computational complexity**, potentially

limiting real-time applications. The **dependence on color information**, particularly for initial segmentation, posed challenges in cases of color fading or adverse conditions. Additionally, while invariant geometric moments effectively handled standard shapes, they may have **struggled with non-standard or irregular signs**, and the system's sensitivity to partial occlusions could have affected performance.

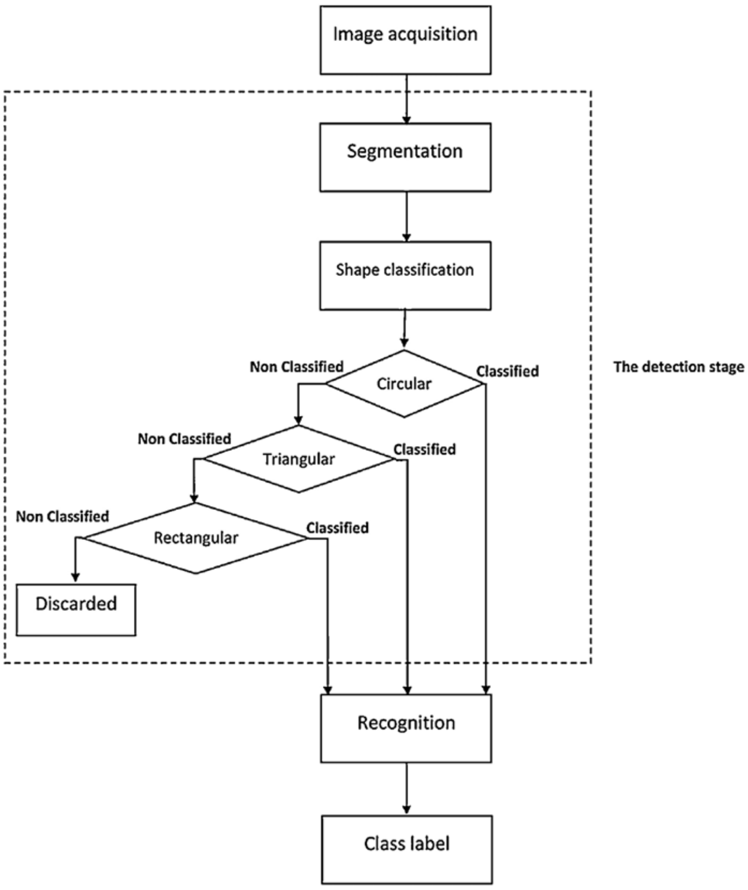

Figure 2.2.3 Block diagram of the proposed method in [3].

2.2.2 Review on Circular Shape Detection

Done by: Cheng Puei Ying

To detect road signs more accurately using shape, [4] focused on innovative method for the **detection and recognition of circular traffic sign**. The detection approach used a new circle detection algorithm paired with an RGB-based color thresholding technique. For recognition, [4] introduced an ensemble of features, including histogram of oriented gradients, local binary patterns, and Gabor features, all within a SVM classification framework.

First of all, [4] incorporated the EDCircles algorithm, which refined the basic circle detection approach in their system. It was an advanced technique that improved upon traditional methods by integrating edge detection and circle fitting algorithms. This approach enhanced the algorithm's ability to detect circles with greater precision, even in challenging conditions such as partial occlusions or varying lighting. The processes of EDCircles first started by detecting edges in the image, followed by extracting the circular arcs from the edge segments, and lastly combining the arcs to generate a circle candidate. These steps helped in accurately identifying circular traffic signs that might be partially obscured or distorted.

```
// I: Input grayscale image
EDCircles(I)
  1. EdgeSegments = DetectEdgeSegmentsBy_EDPF(I);
  2. Arcs = ExtractArcs(EdgeSegments);
  3. CandidateCircles = CombineArcsToCircles(Arcs);
  4. ValidCircles = ValidateCircles(CandidateCircles);
  5. return ValidCircles;
End-EDCircles
```

Figure 2.2.4 Pseudocode of EDCircles.

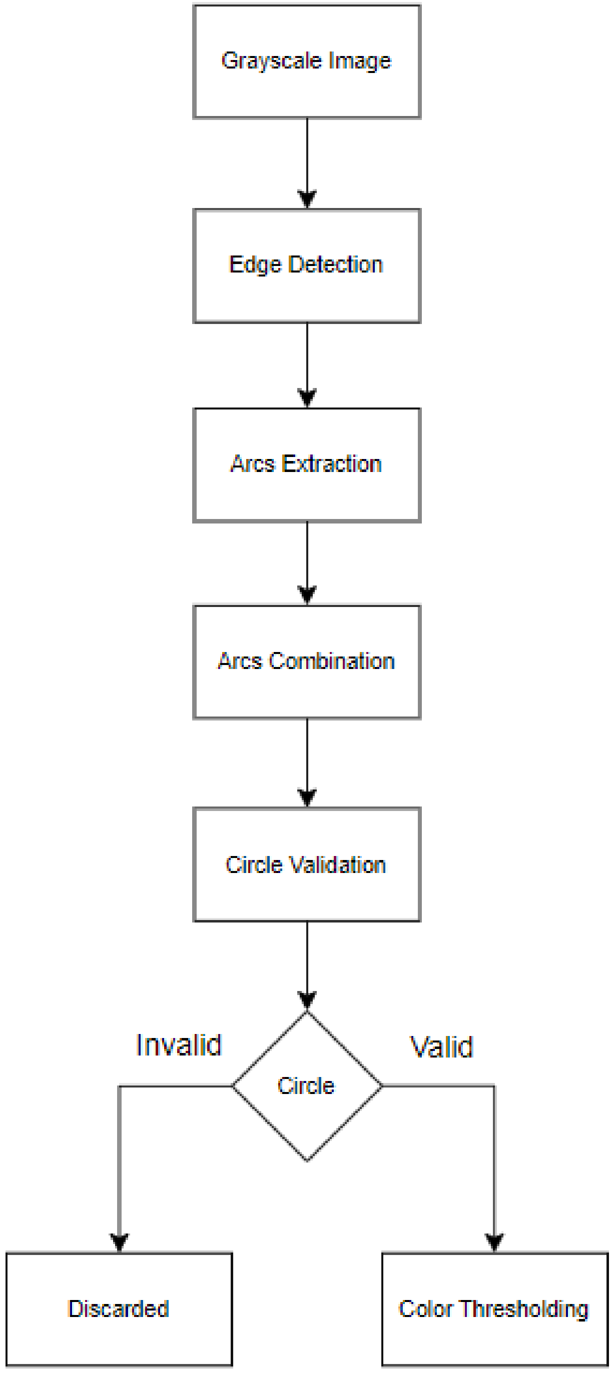

Figure 2.2.5 Block diagram of EDCircles.

Following the circle detection phase, [4] used an **RGB-based color thresholding** technique to further refine the detection process. This step involved analyzing the color components of the image to filter out non-relevant regions and focus on areas likely to contain traffic signs. The RGB-based technique applied color thresholds that matched the typical colors of traffic signs, such as red, blue, and yellow. By isolating regions of interest based on their color, the system reduced the amount of image data that needed to be processed, thus improving efficiency and accuracy. This method helped to validate whether the circle candidate is a real circular traffic sign or just a circular non-traffic sign object.

$$\mathrm{Red}(i,j) = \begin{cases} True,\ if r(i,j) \geq g(i,j)\ and\ r(i,j) \geq b(i,j) \\ \quad and\ g(i,j)/|r(i,j) - g(i,j)| \leq ThR_1 \\ False,\ otherwise \end{cases}$$

$$Blue(i,j) = \begin{cases} True,\ if b(i,j) \geq r(i,j)\ and\ g(i,j)/b(i,j) \leq ThB_1 \\ False,\ otherwise \end{cases}$$

Figure 2.2.6 Equations used in [4] for red and blue color thresholding.

Moving to the feature extraction phase, [4] employed a combination of HOG, LBP, and Gabor features to capture various aspects of the traffic signs. HOG features were used to **capture the gradient information** and **directional intensity variations** in the image, which were useful for identifying the shapes and structures of traffic signs. LBP features **provided texture informatio**n, helping to differentiate signs based on their surface patterns. Gabor features added another layer of detail by **capturing frequency information and texture patterns**, which was particularly useful for recognizing complex sign designs. The integration of these features ensured a comprehensive representation of each traffic sign, enhancing the accuracy of the recognition process.

The extracted features were then classified using a SVM, a powerful classifier known for its effectiveness in handling high-dimensional data. The SVM processed the HOG, LBP, and Gabor features to determine the most likely category for each detected traffic sign. Trained on a large dataset of labeled signs, the SVM learned to distinguish between different types of signs based on the features extracted. It created a decision boundary (hyperplane) that separated the classes, ensuring accurate classification. This approach was particularly effective in managing the complexity of the feature set and providing reliable recognition results.

In conclusion, Paper [4] introduced a robust method for detecting and recognizing circular traffic signs using an advanced circle detection algorithm, **EDCircles**, which improved precision by integrating edge detection and circle fitting. The approach also included **RGB-based color thresholding** for better refinement and feature extraction using **HOG, LBP, and Gabor features**, all classified by a **SVM** for accurate recognition. However, the method **reliance on circle detection** may limit its effectiveness for non-circular signs, and the RGB-based technique might struggle with **faded or poorly illuminated signs**. Additionally, the complexity of feature extraction and classification could **challenge real-time applications**.

2.3 Review on TSDR Methods

Done by: Kok Yung Thow

According to [5], traffic signs detection and recognition (TSDR) system procedure were categorized into three phases, including detection, tracking and classification. Detection stage involves segmentation of localized traffic signs into ROIs from scene image input whereas classification stage refers to the identification and interpretation of traffic signs from ROI candidates based on the extracted features. In some literature, traffic signs recognition also considered as detection and classification, while tracking stage is the procedure of recognition performance enhancement.

Detection Stage

Traffic signs detection (TSD) methods have been generally classified into three approaches: colour based, shape based and the combination of them.

Colour-based TSD methods is one of the first approaches in detecting traffic signs that utilizes colour information of traffic signs that coloured in visible contrasting colours and transform them into different colour spaces. Its characteristics of requiring low computing requirements and effectiveness is the main advantages. However, it is not reliable as the colours will be affected by the weather condition, poor road signs condition and other factors, resulting the failure of localizing traffic signs. Colour based TSD methods mainly used as image pre-processing in TSDR system in recent years. The method types of colour-based techniques, published papers, implemented colour spaces and methods, the advantages, as well as the corresponding detected road sign categories, number of images begin tested and the image type were summarized in [5], Figure 2.3.1, based on their popularity.

Techniques	Paper	Segmentation Methods	Advantages	Sign Type	No. of Test Images	Test Image Type
Color Thresholding Segmentation	[37]	RGB color segmentation	Simple	Any color	2000	N/A
	[38]	RGB color segmentation with enhancement of color	Fast and high detection rate	Red, blue, yellow	135	Video data
HSI/HSV Transform	[40]	HSI thresholding with addition for white signs	Segments adversely illuminated signs	Any color	N/A	High-res
	[33]	HSI color-based segmentation	Simple and fast	Any color	N/A	N/A
	[41]	RGB to HSI transformation	Segments adversely illuminated signs	Any color	N/A	Low-res
	[42]	RGB to HSI transformation	N/A	Red	N/A	Low-res
	[43]	RGB to HSI transformation	N/A	Any color	3028	Low-res
	[44]	HSI color-based segmentation	Simple and high accuracy rate	Red, blue	N/A	Video data
	[45]	HSI color-based segmentation	Simple and real time application	Any color	632	High-res
Region Growing	[48]	Started with seed and expand to group pixels with similar affinity	N/A	N/A	N/A	N/A
	[47]			N/A	N/A	High-res
Color Indexing	[50]	Comparison of two any-color images is done by comparing their color histogram	Straightforward, fast method	Any color	N/A	Low-res
	[49]			Any color	N/A	N/A
Dynamic Pixel Aggregation	[52]	Dynamic threshold in pixel aggregation on HSV color space	Hue instability reduced	Any color	620	Low-res
CIECAM97 Model	[54]	RGB to CIE XYZ transformation, then to LCH space using CIECAM97 model	Invariant in different lighting conditions	Red, blue	N/A	N/A
YCbCr Color Space	[55]	RGB to YCbCr transformation then dynamic thresholding is performed in Cr component to extract red object	Simple and high accuracy	Red	193	N/A
	[56]		High accuracy less processing time	Any color	N/A	Low-res

Figure 2.3.1: Examples of colour based TSD methods summarized in [5].

Shape-based TSD methods detect the traffic signs with their shapes. It has been developed based on the shape characteristics of road signs such as circle, triangle, rectangle, and octagon. Traffic signs detected by their shape outline without using the colour information. Shape based approach immunes the colour changes due to the external factors and increases the ROIs capturing efficiency. Unfortunately, these methods require high computational power when tracking the traffic signs outline in large images and they are not performed well for traffic signs with poor conditions and visibility. [5] summarized the shape based TSD methods, as shown in Figure 2.3.2, according to their names, proposed papers, overall processes, recognition features as well as the sign type.

Technique	Paper	Overall Process	Recognition Feature	Advantages	Sign Type
Hough Transform	[77]	Each pixel of edge image votes for the object center at object boundary	N/A	Invariant to in-plane rotation and viewing angle	Octagon, square, triangle
	[78]		AdaBoost	High accuracy	Any sign
	[79]		N/A	Robustness to illumination, scale, pose, viewpoint change and even partial occlusion	Red (circular), blue (square)
	[80]		N/A	Reducing memory consumption and increasing utilization Hough-based SVM	Any sign
	[81]		N/A	Robustness	Red (circular)
	[59]		Random Forest	Improve efficiency of K-d tree, random forest and SVM	Triangular and circular
	[82]		SIFT and SURF based MLP	Applying another state refinement	Red circular
Similarity Detection	[52]	Computes a region and sets binary samples for representing each traffic sign shape.	NN	Straight forward method	Any color
DTM	[61]	Capturing object shape by template hierarchy.	RBF Network	Detects objects of arbitrary shape	Circular and triangular
Edge Detection Feature	[63]	A set of connected curves is found which indicates the boundaries of objects within the image.	Geometric matching	Invariant in translation, rotation and scaling	Any color
	[64]		Normalized cross correlation	Reliability and high accuracy in real time	Speed limit sign
	[65]		N/A	Improved accuracy by training negative sample	Red (circular)
	[66]		N/A	Invariant in noise and lighting	Triangle, circular
	[67]		CDT	Invariant in noise and illumination	Red, blue, yellow
Edges with Haar-like Features	[69]	Sums three pixel intensities and calculates the difference of sums by Haar-like features	CDT	Smoother and noise invariant	Rectangular, any color
	[70]		SVM	Fast method	Circular, triangular upside-down, rectangle and diamond

Figure 2.3.2: Table of shape based TSD methods with fewer columns in [5]

In recent years, combination of colour and shape-based TSD methods were implemented to improve the TSDR efficiency. These hybrid methods work with considering colour features after the shape-based approach being implemented, vice versa, or either of them chosen as the main method and integrated with some aspects from another approach[5].

Tracking Stage

Tracking is the procedure of verifying traffic sign and eliminates the redundant detection on the same signs to significantly reduce the computational time. It is crucial for real-time TSDRs in autonomous driving. The process involves fixed camera recorded video frames that are feed into TSDR, then the incorrect sign candidate and those only shows up one once in a time interval will be eliminated. The block diagram to illustrate this concept is shown below in Figure 2.3.3. The most adapted tracking technique was Kalman Filter [5].

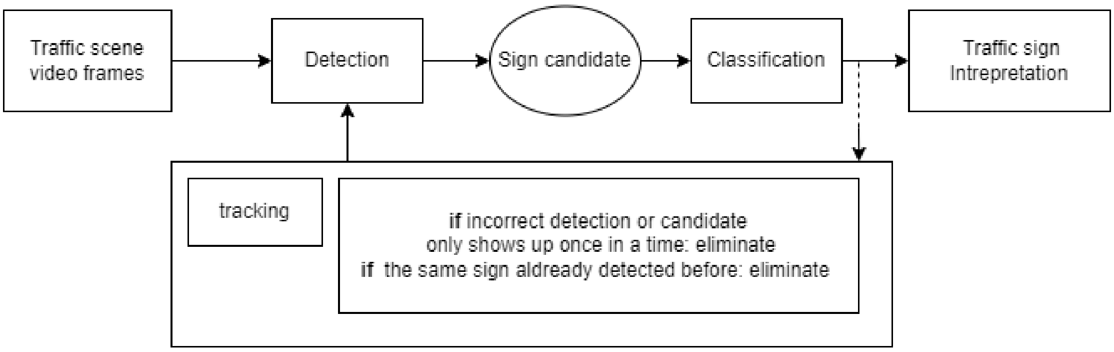

Figure 2.3.3: Block diagram illustrates tracking process adapted from [5]

<u>Classification Stage</u>

Classification refers to the process of interpretation of the meaning or rule of detected traffic signs by applying classifier algorithms. Figure 2.3.4 shows the classification methods summarized in [5].

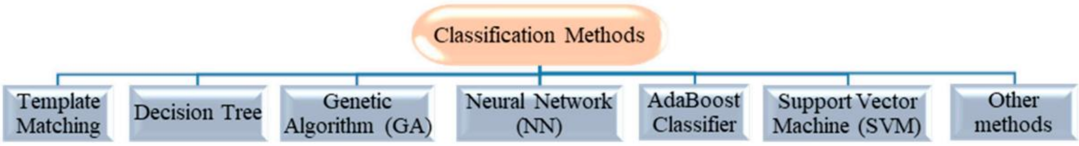

Figure 2.3.4: Conventional Classification Methods concluded in [5]

Template matching is a method that reported with high accuracy and high efficiency that perform small window matching in image searching by utilizing pre-defined templates. Its drawbacks evaluated as low performance in classifying traffic sign images with noise and occlusions due to the high sensitivity. **Decision tree, or random forest** is another classifier that selecting the most frequent output among the individual trees' predictions. It was reported with low computational time needed and good performance. However, this algorithm can be slow due to its complexity and reviewed as not suitable for real-time traffic signs recognition.

Genetic algorithm (GA) was widely used in TSDR in the beginning of this century because its ability to recognize the traffic sign content with shape loss, but due to its high uncertainty in work time and in finding the best solution, it is not popular today. **Artificial intelligence neural network (ANN)** method is one of the most common approaches used in classification process with simultaneous objects recognition with high accuracy rate, adaptive to environments. Unfortunately, the training time can be slow and unstable since it needs a large amount of training data for real world situations.

Adaptive boosting (AdaBoost) is another approach in solving TSDR classification by combining multiple learning algorithms. Its advantages reported as easy to be implemented, high accuracy rate and flexibility in performance improvement. In contrast, the training time needed increases when the input data has a broad variety. Moreover, the models need to be retrained every time when new training samples are updated. **Support vector machine (SVM)** is a classifier that performing separation of data into two categories. It was applied in TSDR classification and was reported as an extremely fast and high accuracy. However, the main advantage of this method is the absence of result transparency, in other words, interpretation on the results is impossible due to high dimensional training data.

Other methods reviewed in [5] in classification task include SIFT matching for recognizing the road sign broken parts, Karhunen-Loeve transform and MLP with low processing time, as well as self-organizing map (SOM) with extremely high accuracy rate.

According to TSD methods review in [6], **light detection and ranging (LiDAR)** has been a popular solution for ADAS and Auto Driving System (ADS) with its mobile mapping technology. It can be used to detect 3D urban objects and applied in inventory analysis and routine inspections with the 3D geometric information detected. Besides, TSD can be done using the combination of 3D information and RGB cameras that collect colour, shape texture information in promising the direction of traffic sign hence support the TSDR tasks. Figure 2.3.5 shows the structure of LiDAR and RGB camera based TSDR system reviewed in [6].

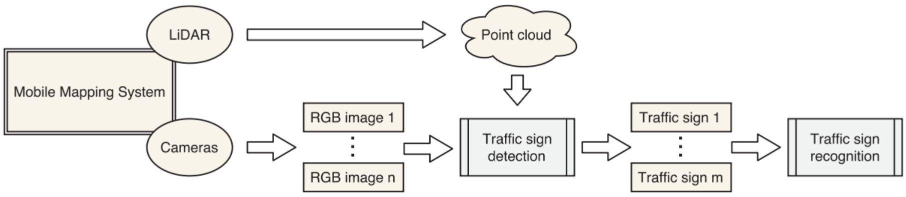

Figure 2.3.5: Block diagram of 3D point cloud and 2D imagery based TSDR system reviewed in [6].

This structure was adapted in detecting multi-view TSDR based on template matching approach and in deep Boltzmann machine (DBM) based hierarchical classifier in deep learning was used in classification stage. Besides, 3D cloud point and 2D imagery method was implemented in two-stage TSDR system, where a **deep neural network (DNN)** was used in classification on RGB images projected from 3D cloud data. However, its low generalization ability [5] and cannot adapt with most of the deep learning methods are the main drawbacks of LiDAR methods.

2.4 Review of Traffic Sign Detection

Done by: Tan Pei Shi

Convolutional Neural Network with Template-Based Method

Recently, traffic sign detection has gained increasing attention in the field of intelligent vehicle navigation due to the distinctive and recognisable characteristics of traffic signs. Their standardised appearance in terms of shape, colour and pattern makes them valuable landmarks for mapping and localisation. This characteristic has been efficiently exploited in [7], where a novel traffic sign detection system was proposed to simultaneously estimate the position and boundaries of traffic signs using a single convolutional neural network (CNN). This CNN was adapted to predict 2D pose and shape labels of planar targets based on template images stored in the shape database, thereby improving the efficiency of the detection process.

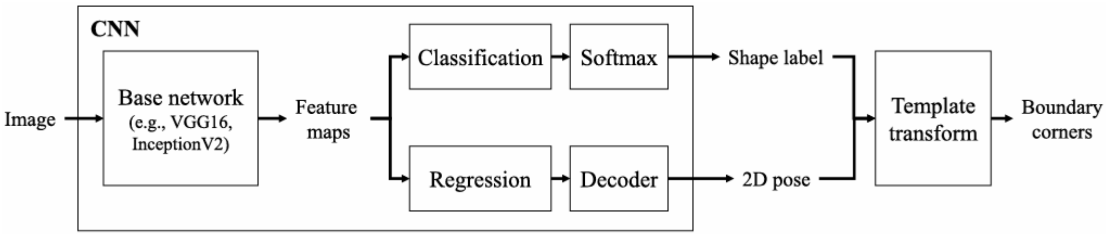

Figure 2.4.1. Block diagram of the proposed method in [7].

As illustrated in Figure 2.4.1, the proposed method began by inputting an image to the CNN block, which employed a base network with an accuracy or speed trade-off, such as VGG16 or Inception V2, to extract the feature maps. These feature maps were then processed simultaneously in two separate convolutional layers, namely the shape classification layer and the pose regression layer. In the shape classification layer, a Softmax function was used to classify the feature maps into several shape categories, resulting in a shape label. Simultaneously, the 2D poses of the traffic signs were estimated in the pose regression layer. The estimated 2D poses and shape classification were then further refined by the decoder.

Finally, the resulting shape label and 2D poses were used for template transformation. This process involved the retrieval of a template image that corresponded to the predicted shape label from the shape database and its projection onto the observed traffic signs, thus allowing for the accurate computation of the boundary corners. The utilisation of template images, as proposed in this method, provided prior insight into the target shapes, thereby improving the accuracy of occluded sign detection. However, it should be noted that the utilisation of template images could also result in a degree of bias, making the system less adaptable to traffic signs with novel or highly deformed shapes that are not represented in the shape database.

To address this issue, our project proposed the use of a hybrid approach that combines the template-based method with a multiscale attention mechanism, where capturing features at multiple levels can help in focusing on important features [8]. This combination is expected to reduce the bias inherent in template-based methods while increasing the adaptability of the system. The CNN with multiscale attention enables the focus on the most relevant features at different scales, whereas the template-based methods facilitate precision in shape classification and boundary detection.

Cascaded R-CNN with Multiscale Attention and Imbalanced Samples

There are a variety of factors contributing to poor sign detection efficiency (e.g., non-detection and false detection). These factors included the small size of traffic signs, adverse weather conditions, changes in lighting, and the resemblance of signs to real traffic signs. In order to address these challenges, the authors of [8] proposed a cascaded R-CNN with multiscale attention and unbalanced samples for traffic sign detection, as shown in Figure 2.4.2.

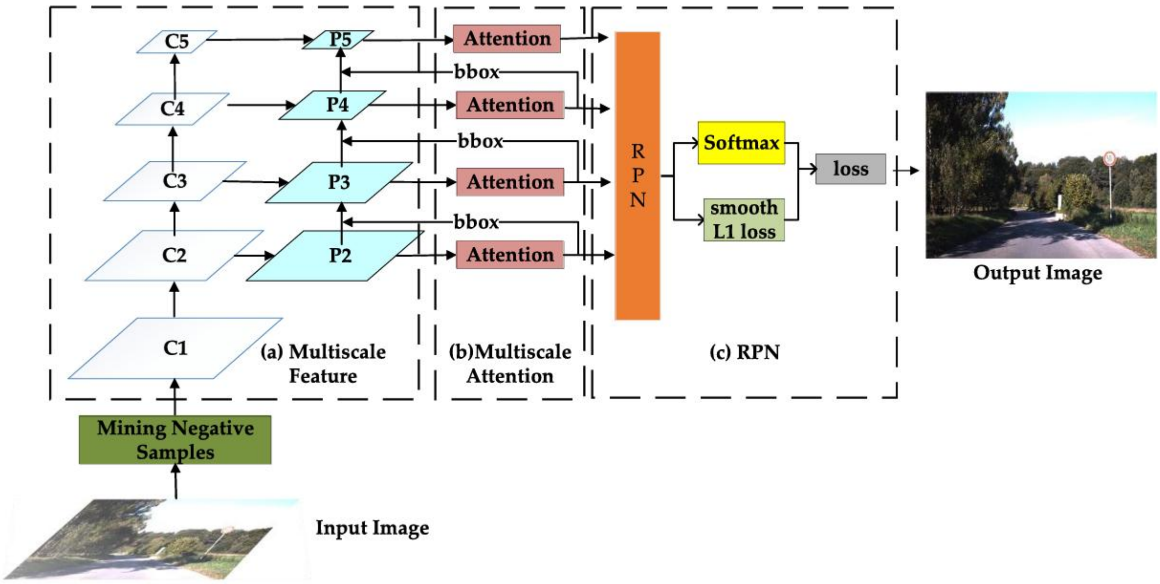

Figure 2.4.2. Block diagram of the proposed approach in [8].

According to Figure 2.4.2, the process began with an input image, which might or might not contain a real traffic sign, being passed into the model. The model first processed the input image to identify any negative samples (false positives) within it, where negative samples contained regions that did not contain traffic signs but were misclassified by the model as containing traffic signs. These identified hard negatives were used to refine the model, enhancing its capability to distinguish true traffic signs without being tricked by signs that look like true traffic signs.

Once hard negatives were identified, the image was processed through a backbone network, namely ResNet50, where feature maps (C1 to C5) were extracted at five different scales. These feature maps were then processed by the feature pyramid network, resulting in multiscale feature maps (P2 to P5). In the multiscale attention model (as shown in Figure 2.4.3), dot product and Softmax operations were performed on the multiscale feature maps layer by layer to obtain a weighted multiscale feature. The weighted feature was then used to generate a bounding box (bbox) that proposed regions of interest (ROI). In order to enable the model to focus on the ROI more efficiently while discarding irrelevant areas, the generated bounding box was fused with the next feature map (e.g., the bounding box resulting from P2 was fused with P3). These refined feature maps were then passed back into the multiscale attention module again, where they underwent further refinement, thereby enhancing the accuracy of traffic sign detection through the iterative application of refined features.

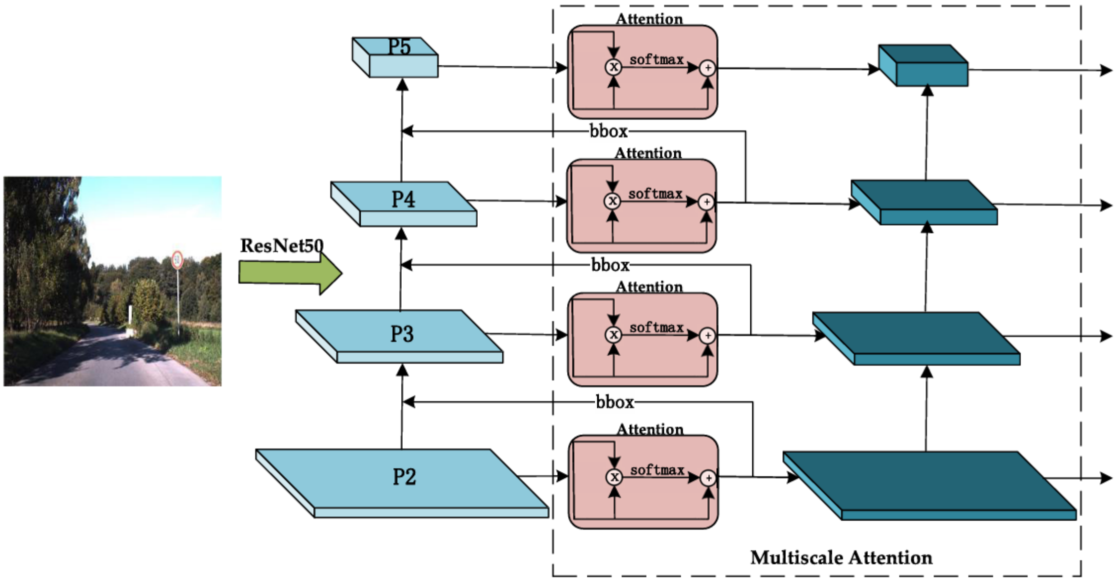

Figure 2.4.3. More detailed version of the block diagram for the multiscale attention part in [8].

After that, all refined feature maps derived from multiscale attention were then used as inputs to the Region Proposal Network (RPN) to generate region proposals. These region proposals were then classified as either traffic signs or background using the Softmax function (classification loss), while the Smooth L1 loss was used to refine the prediction of the bounding boxes (localisation loss). Both outputs from the Softmax function and Smooth L1 loss were combined in the loss function to produce an output image in which traffic signs were correctly identified and localised.

Although the approach proposed in [8] can effectively solve the factors of poor sign detection, such as the small size of traffic signs, lighting and weather changes, it has also increased the computational complexity, since the proposed method consists of multiple stages. Our project addresses this issue by proposing the use of a hybrid approach, as mentioned above. With the use of template images to identify the potential shape of traffic signs, the computational complexity in the multiscale attention can be reduced by focusing on the potential regions.

2.5 Comparison of the Four Techniques

Done by: Every group member

Ang Jia Pei	Done by	Cheng Puei Ying
a. Color Invariants (Gaussian Color Model) [1] b. Pyramid Histogram of Oriented Gradients (PHOG) [1] c. HOG-Based Method with SVM [2]	**Technique**	Invariant Geometric Moments [3]
a. <u>Gaussian Color Model</u> By translate the RGB color model to Gaussian color model to segment images to identify candidate regions of interest (ROIs) and isolate traffic signs from	**Explanation**	A set of scalar values that provide a summary of the shape's geometry. For a given shape, these moments can be calculated to describe various properties such as area, centroid, and orientation.

complex backgrounds.		
b. Pyramid Histogram of Oriented Gradients (PHOG) Canny edge detector is used to obtain edge contours, which are then divided into cells at different pyramid levels, and all the HOG vectors concatenated and create the final PHOG descriptor.		
c. HOG-Based Method with SVM Incorporates colour information into HOG		

<table>
<tr><td>algorithms by using SVM for better accuracy and handle complex background.</td><td></td><td></td></tr>
<tr><td>a. <u>Gaussian Color Model</u>
Linear translation from RGB to Gaussian model, K-means clustering, hill-climbing algorithm for K-seeds.</td><td rowspan="3">Features</td><td rowspan="3">Used for shape detection of traffic signs by capturing essential shape features that are independent of scale, rotation, and translation. Focuses on shape classification by matching segmented Regions of Interest (ROIs) with known shapes such as circles, triangles, rectangles, and octagons.</td></tr>
<tr><td>b. <u>Pyramid Histogram of Oriented Gradients (PHOG)</u>
Chromatic-Edge (improved PHOG) with Otsu method is applied to filter out weak edges and reducing noises.</td></tr>
<tr><td>c. <u>HOG-Based Method with SVM</u></td></tr>
</table>

Integrates color data into HOG descriptors and uses SVM training iteratively to improved accuracy and background handling.		
a. <u>Gaussian Color Model</u> Used for color-based image segmentation and noise reduction by modeling color distributions as Gaussian distributions.		• After segmentation, the system identifies the geometric shapes within the ROIs. The focus is on shapes commonly associated with traffic signs, such as circles, triangles, rectangles, and octagons.
b. <u>Pyramid Histogram of Oriented Gradients (PHOG)</u> The improved PHOG descriptor is obtained by concatenating all HOG vectors at each pyramid resolution.	**Usage**	• The system calculates the invariant moments for each segmented region and compares these moments against pre-calculated moments for standard traffic sign shapes. • The matching process is
c. <u>HOG-Based Method with SVM</u>		straightforward: the system compares the invariant

<table>
<tr><td>Enhances traffic sign detection by combining color information with HOG features and applying iterative SVM training to manage background variations.</td><td></td><td>moments of the detected shape with those of known traffic sign shapes. If there is a close match, the shape is classified accordingly.</td></tr>
<tr><td>a. <u>Gaussian Color Model</u>
Improves recognition under different lighting conditions and weather.</td><td rowspan="2">Advantages</td><td rowspan="2"> Reduced Need for Training Data: Unlike machine learning models, which typically require extensive training data, this technique relies on pre-defined shape descriptors. This can significantly reduce the development time and computational resources needed.
 Efficiency: The computation of invariant moments is relatively efficient, making the technique suitable for systems</td></tr>
<tr><td>b. <u>Pyramid Histogram of Oriented Gradients (PHOG)</u>
• Filter weak edges and reducing noise
• Improved recognition</td></tr>
</table>

through spatial pyramid structure		where processing speed is important, such as in real-time traffic sign recognition for autonomous vehicles.
c. <u>HOG-Based Method with SVM</u> Enhances detection accuracy and reduces memory usage while managing background variations effectively		□ **Simplified Detection Process:** By directly matching shapes using invariant moments, the method bypasses the complexity of model training and tuning, which is often required in machine learning approaches.
a. <u>Gaussian Color Model</u> • Distort perceived colors of an object, affecting accuracy • Sensitive to complex backgrounds and overlapping colors	**Disadvantages**	□ **Sensitivity to Occlusions:** Although invariant to geometric transformations, this method can struggle with partial occlusions, where only a part of the sign is visible. This could lead to incorrect shape classification or missed detections. □ **Dependency on Segmentation:** The accuracy

		of the invariant moment calculation heavily depends on the quality of the initial segmentation. If the color thresholding step fails to accurately segment the sign from the background, the subsequent shape detection might be flawed. **Non-Standard Shapes:** The method is well-suited for detecting standard geometric shapes but may have difficulty with more complex or non-standard traffic sign shapes. These might require additional processing steps or alternative detection methods.
b. <u>Pyramid Histogram of Oriented Gradients (PHOG)</u> • Perform poorly when encounters		

blurred or rotated traffic signs • Decrease in the rate of accuracy in complex scenes with overlapping colors or textures		
c. HOG-Based Method with SVM Higher computational complexity and increased feature vector dimensionality		

Table 2.5.1 Comparison between Colour Invariants (Gaussian Color Model) [1], Pyramid Histogram of Oriented Gradients (PHOG) [1], HOG-Based Method with SVM [2], and Invariant Geometric Moments [3].

Kok Yung Thow	**Done by**	**Tan Pei Shi**
LiDAR & RGB with DNN	**Technique**	• CNN with Template-Based Method [7] • Cascaded R-CNN with Multiscale Attention and Imbalanced Samples [8]
	Features	**Common Features**

3D cloud points & 2D imagery applied in detection and DNN as TDSR classifier		• Involved CNN-based feature extraction.
		Specific Features
		CNN with Templated-Based Method [7]: • Employed a decoder to refine the estimation of 2D poses and shape classification. • Performed template transformation to accurately compute boundary corners.
		Cascaded R-CNN with Multiscale Attention and Imbalanced Samples [8]: • Employed a multiscale attention mechanism for focusing on important features. • Involved hard negative mining to improve the robustness of the model.

Extremely high accuracy	**Advantages**	CNN with Templated-Based Method [7]: • Able to predict the shape and position of traffic signs efficiently. • Improve the accuracy of occluded sign detection.
		Cascaded R-CNN with Multiscale Attention and Imbalanced Samples [8]: • Able to detect traffic signs in various situations, such as signs of small size, adverse weather conditions or changes in lighting effectively. • Reduce false detections.
Low generalization ability	**Disadvantages**	CNN with Templated-Based Method [7]: • Less adaptable to traffic signs with novel or highly deformed shapes that are not represented in the shape database.

		Cascaded R-CNN with Multiscale Attention and Imbalanced Samples [8]: • Lead to higher computational complexity due to the involvement of multi-stages.

Table 2.5.2 Comparison between LiDAR, RGB with DNN, CNN with Template-Based Method [7] and Cascaded R-CNN with Multiscale Attention and Imbalanced Samples [8].

CHAPTER 3 - Proposed Method/Approach

3.1 Design Specifications

3.1.1 Methodologies and General Work Procedures

Done by: Cheng Puei Ying

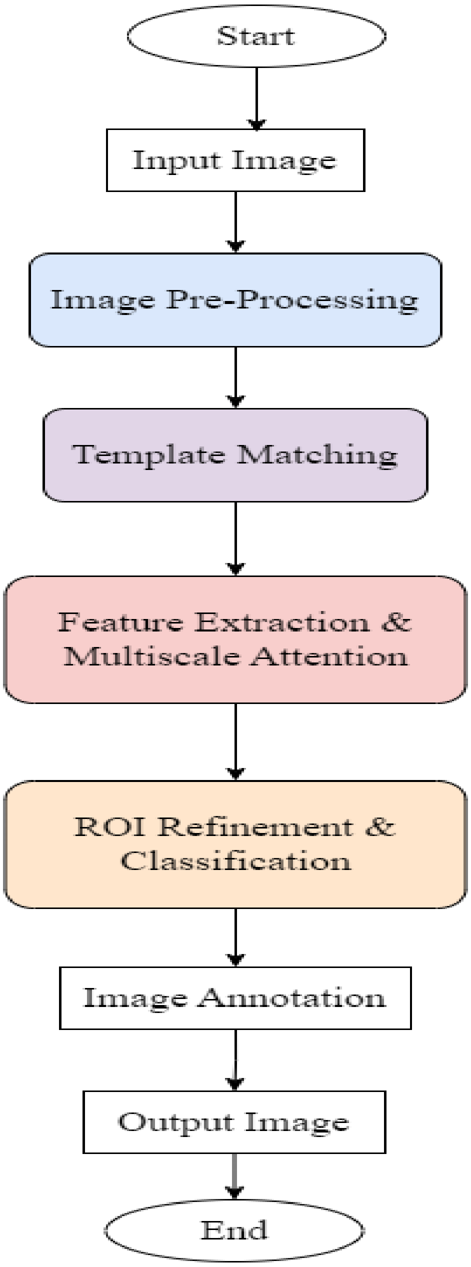

Figure 3.1.1 System Block Diagram

First of all, the user is required to input images captured by camera installed in the vehicle. This step is considered as **image acquisition**.

Next, the input images will undergo **pre-processing** so that they are more suitable for detection later. Since the input images are in the RGB color space as it closely aligns with how human vision perceives, the images are first converted it to **HSI color space**. Each image pixel is classified according to its hue, saturation, and intensity. This separation helps in distinguishing objects based on color, regardless of lighting conditions.

To further improve the quality of the input image, **Gaussian filter** is then applied to the converted images to smooth them and reduce high-frequency noise. As a result, the images become neater and less susceptible to random variations that interfere with further processing.

Following the pre-processing, the images proceed to the **template matching** stage. In this step, the system uses predefined templates that represent different traffic signs. The processed images are scanned to find regions that closely match these templates, thereby isolating potential areas of interest that might contain traffic signs.

Next, in the **feature extraction and multiscale attention** phase, the system extracts relevant features from the identified regions. This step involves analyzing various aspects of the regions, such as edges, corners, and textures, which are crucial for accurate recognition. Multiscale attention is applied to focus on these features at different scales, ensuring that both large and small signs are correctly identified.

The following step is **ROI refinement and classification**. After isolating the regions of interest in the previous steps, the system analyzes these regions to identify and classify the shapes of the detected traffic signs like circles, triangles, rectangles, and octagons.

Lastly, the images undergo **image annotation**. The system extracts the detected regions that correspond to traffic signs from the image. These regions are then used for further analysis and classification, ensuring that only the relevant parts of the image are processed for accurate traffic sign recognition.

3.1.2 Tools to Use

Done by: Cheng Puei Ying

Hardware used

1. Laptop

Brand	MSI Modern 15 B12M-064
Operating System	Windows 11
Processor	Intel® Core i7-1255U
RAM	16GB DDR4 3200MHz
GPU	Intel Iris Xe Graphics
Hard Disk	512GB NVMe PCle Gen3x4 SSD

Table 3.1.1 Laptop Specification

Software used

[1] Microsoft Visual Studio 2022

Version: 17.10.5

[2] OpenCV

Version: 4.1.0

3.1.3 User Requirements

Done by: Cheng Puei Ying

· User shall provide color images of traffic signs for detection.

· User shall ensure that the inputted images contain only traffic signs or relevant road scenes. ??

· User shall receive the detected traffic sign types after the recognition process.

· User shall observe the characteristics and details of the traffic signs based on the inputted images. ??

3.1.4 System Performance Definition

Done by: Cheng Puei Ying

Don't seem like the correct contents like real time, accuracy of how many %, …

Since traffic road sign detection and recognition play a crucial role for autonomous vehicles and system failures can have serious consequences, it is important to ensure that the system operates efficiently and without errors.

Firstly, during the image preprocessing phase, color space conversion plays a vital role. The input images, initially captured in the **RGB color space, are converted to the HSI color space**. This approach makes it more effective for distinguishing objects based on color under varying lighting conditions. This also helps in improving the accuracy of detecting traffic signs, as the system can better focus on the color characteristics that are relevant to traffic signs.

Next, **Gaussian blurring** is applied to the converted images to reduce high-frequency noise and smooth the images. This noise reduction technique enhances the quality of the input images by minimizing random variations that could interfere with further processing. By smoothing the images, the system can more effectively detect and recognize traffic signs, leading to improved accuracy in the final outcome.

Following pre-processing, the system employs **template matching** as the next critical step. This method involves comparing regions within the pre-processed images to predefined templates that represent various traffic signs. By aligning the image regions with these templates, the system can accurately identify areas that potentially contain traffic signs, setting the stage for more detailed analysis.

Once potential traffic signs are identified, the system proceeds with **feature extraction and multiscale attention**. During this phase, the system extracts crucial features from the identified regions, such as edges, shapes, and textures, which are key to distinguishing one traffic sign from another. The use of multiscale attention allows the system to focus on features at different scales, ensuring that both large and small signs are detected and recognized with precision.

Next, **ROI refinement and classification** takes place, where the ROI identified through feature extraction are refined and classified. The system automatically identifies the shapes of traffic signs, whether they are circles, triangles, rectangles, or octagons, and classifies them accordingly. This step is fully automated, enhancing the efficiency of the detection process and ensuring that the correct traffic sign is recognized without the need for manual intervention.

After classification, the system performs **image annotation**, where the detected and classified traffic signs are marked directly on the images. This annotation provides a clear and visual representation of the system's findings, making it easier to understand and analyze the detected signs. Finally, the system outputs the annotated images, completing the detection and recognition process.

The system performance is defined by several critical metrics. First of all, **accuracy** measures the system's capability to correctly identify and classify traffic signs. The system is able to achieve a high proportion of true positives compared to the total number of actual signs. In addition, the system should also have **short processing time**, allowing it to handle and analyze images quickly, ensuring real-time or near-real-time operation. This enables it to make decision faster during operation. By addressing these aspects, the system aims to provide reliable, efficient, and practical traffic sign detection to enhance driver assistance and autonomous driving capabilities.

3.1.5 Verification Plan

Done by: Cheng Puei Ying

The proposed traffic sign detection system aims to **achieve high accuracy and consistency** in detecting and classifying traffic signs in real-world images. However, several factors could potentially compromise system performance, given the challenges inherent in real-life environments. These challenges include **low-resolution images**, **variations in lighting and contrast**, **occlusions**, **complex backgrounds**, **distortions**, and so on that could lead to misclassification or false detections.

Specific scenarios that could affect system performance include:

- The traffic sign images may be captured from various angles, leading to slight distortions or rotations that could challenge shape recognition algorithms.
- Variations in lighting conditions, such as overexposure or shadows, could alter the appearance of traffic signs, making them harder to detect and classify accurately.
- Some traffic signs might be partially obscured by objects like foliage, vehicles, or other obstructions, potentially leading to incomplete or incorrect detection.

These challenges must be addressed to ensure that the system remains robust and reliable across diverse real-world conditions.

3.2 System Design / Overview

Done by: Tan Pei Shi

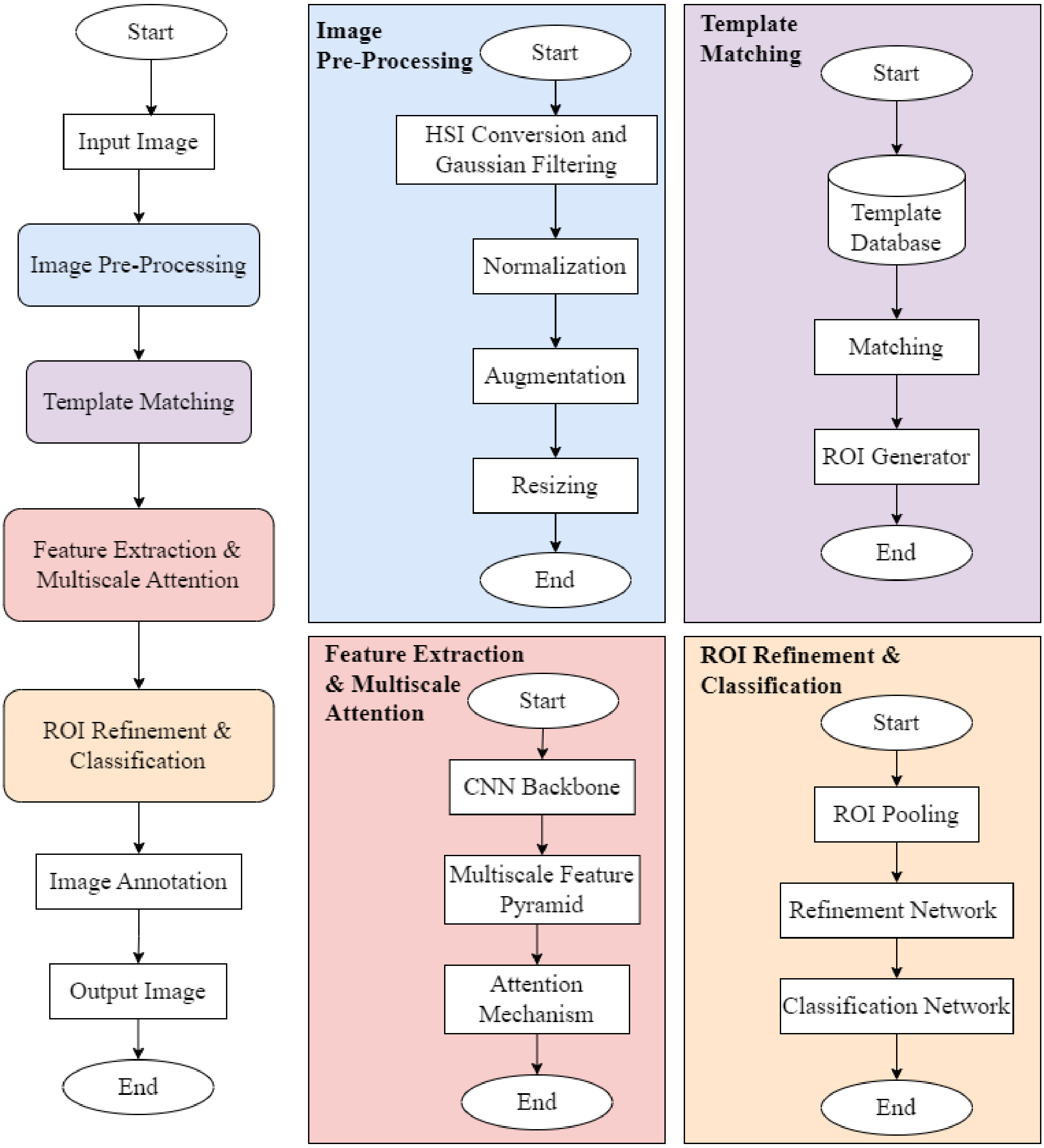

Figure 3.2.1. Block diagram of the system design.

As illustrated in Figure 3.2.1, the system begins with a real-time image captured by the embedded camera of the autonomous vehicle as input data.

3.2.1 Image Pre-Processing

When the captured image is loaded into the system, it will first undergo image pre-processing steps such as HSI conversion and Gaussian filtering, as mentioned in section 3.1.1, to enhance the image quality and colour-based features. After that, normalisation will be performed on the image to ensure that each captured image will have similar feature scales, which can avoid scale differences that may lead to biased results. Once the image is normalised, augmentation techniques such as rotation, flipping, cropping and more will be performed on the image to produce more images with different effects that could be used to train the model for a better performance by avoiding the occurrence of overfitting. Finally, the image will be resized to ensure that each image has a standard size that fits the architecture of the designated model.

3.2.2 Template Matching

Upon image pre-processing, template matching will be performed on the processed image by comparing the shape and colour of templates stored in the template database with the input image. In order to determine the degree of similarity between the template image and the input image, normalised cross-correlation (NCC) is chosen as the scoring metric. Once all the potential regions in the image reach the threshold value of 0.92 for NCC, early termination of template matching will be performed to provide faster detection and reduce the computational cost. Through the matching score, all potential regions of interest (ROIs) within an image can be identified and extracted as bounding boxes for further processing.

3.2.3 Feature Extraction & Multiscale Attention

The ROIs extracted by template matching will then be passed into a ResNet CNN backbone for feature extraction without the need to start from scratch such as applying filtering on the input image. Here is where a hierarchy of feature maps will be generated to represent multi-level abstractions. These feature maps will then be combined to create a multiscale feature pyramid, which allows features to be detected at different scales. In order to improve the accuracy of sign recognition, the extracted ROI will be adjusted to align precisely with the corresponding feature map levels using bilinear interpolation. After that, a multiscale attention mechanism will be applied to weigh the feature maps based on their relevance. These weighted feature maps can then be used for the classification of sign types.

3.2.4 ROI Refinement & Classification

The resulting weighted feature maps will then undergo refinement and classification process. ROI pooling will first be applied to convert all weighted feature maps which consist of variable sizes into fixed-size feature maps. The pooled feature maps will then be processed in the refinement network that performs corner regression to predict the offset needed to adjust each coordinate of bounding boxes independently. Corner regression is selected due to its high flexibility in the adjustment of bounding box shapes. Later, the refined ROI features will be passed into the classification network to undergo classification by utilising a softmax function. Each refined ROI will be categorised into a specific traffic sign category.

When both the bounding boxes and signs category of ROIs are finalised, the system will proceed to the image annotation step, whereby bounding boxes and labels will be localised on the detected traffic signs of the input image. This labelled image will thus serve as the output of the system.

3.3 Implementation Issues and Challenges

Done by: Kok Yung Thow

The major issue of the project implementation is the need of **funding** for equipment to significantly reduce the training time and extend the performance of the proposed real-time TSDR system. A high-resolution camera for car such as REDTIGER F7NP 4K font camera is required in the implementation stage. The reason is to use the camera to collect training samples and then use them in image processing tasks of the proposed TSDR system as illustrated in Figure 3.2.1. This is important to ensure the quality of the learning process of the template database using colour and shape invariants where it will be utilized in template matching. Besides, a high performance external graphic processing unit (eGPU) like Razer Core X Chroma eGPU is needed to extend the hardware performance. This is because high computational power can effectively reduce the implementation time can explore more potential enhancement of our proposed TSDR system. By utilizing the eGPU, the time needed for TSDR training time can be shorten in image processing and deep learning tasks as well as the classification tasks.

In addition, **volunteers** are needed to enhance the project quality in the implementation stage. They will be requested for providing the road scenes that are recorded by the car dash camera when they are driving in daily routine. This is crucial for gaining more training samples with different road scene with external factors such as different road sign conditions, occlusions and illumination problems. Hence, the template database that collects the features of different types of traffic sign can be improved in terms of quality and quantity. This will lead to a real time TSDR development that can be implemented in autonomous driving systems.

3.4 Timeline

Done by: Ang Jia Pei

The Gantt chart in Figure 4 shows a complete timeline for producing the final year project report; it has been segmented into reports 1 and 2 that are crucial for project execution successfully. Spanning from June to September, it outlines the detail planning phase to the implementation phase. In report 1, the planning and analysis phases take place, which include activities such as determining the motivation and problem statement of this project and further proceeding to the review of the research paper to find out the work had been done by other researchers. Then, proceed to the development and implementation phase, which is to design the specification for this project and try to study the result of previous techniques implemented by other researchers. Moving on, after the finalization and submission of report 1, proceed to report 2, detail implementation of the code carried out, and testing on 84 traffic images to get the result. Finally, summarization of report 2 and submission. This visual schedule facilitates the tracking progress clearly, ensuring adherence to deadlines and effective time management throughout the project.

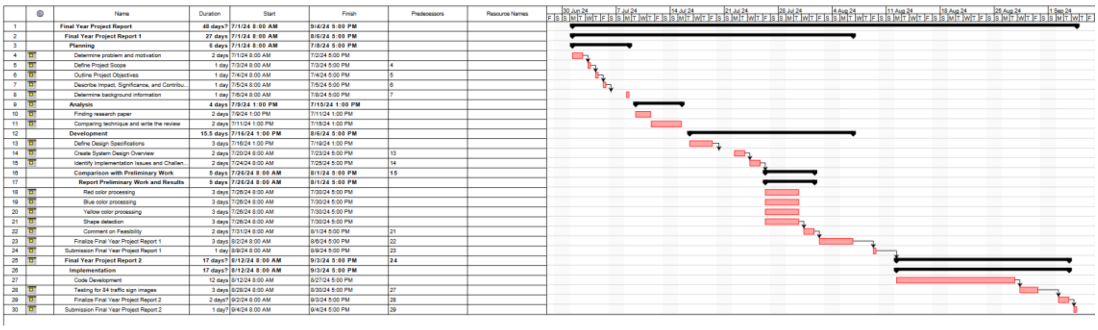

Figure 3.4.1 Gantt Chart for timeline

Milestones chart missing.
Font size too small to see.

CHAPTER 4 - Preliminary Work

4.1 Coding Work 1 & Results (Red Colour)

Done by: Tan Pei Shi

4.1 Coding Work 1 – Explanation (Refer to Appendix 2)

The preliminary work on the segmentation of red colour traffic signs has achieved a 100% segmentation rate, meaning that all signs with red colour have been

Contradicting of 100% vs 3 out of 28?

successfully segmented from the image. However, 3 out of the 28 total samples were unable to be fully segmented out from their backgrounds, which had similar colours to the dominant signs of the image (as shown in Figure 4.1.1). This issue is primarily due to the overreliance on the longest contour during the segmentation process. Since the element with the longest contour is assumed to be the dominant sign, elements adjacent to the actual sign with similar colours may be misidentified (false positive) as part of the traffic sign due to the continuous contour from the actual sign (as shown in Figure 4.1.2). However, this issue may be solved by performing logical AND operations

Cannot see how. More detail on this to be shown later?

between the respective mask of colour and shape segmentation.

Figure 4.1.1 Images where traffic signs were not fully isolated from their background.

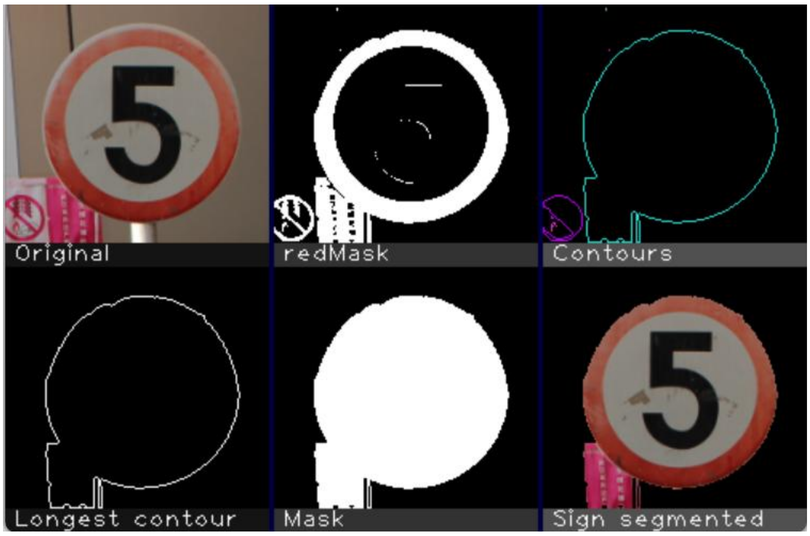

Figure 4.1.2. An image where the adjacent elements were misidentified as part of the actual sign due to the use of the longest contour.

Where to find this? This is only for one example, not for occlusion in general!!

Besides, preliminary work on red-coloured sign segmentation has proven that occluded signs can also be successfully isolated from their background image. However, the obscured part will not be automatically reshaped back to its original form (as shown in Figure 4.1.3). In order to reconstruct the missing portion of the sign, a template-based method may be used to predict and fill the potential shape with its corresponding colour. However, it should be noted that this method is only suitable for traffic signs that are not heavily occluded. No conclusion then.

The concrete contribution here is only the suggestions made?

Need more detail to emphasize your effort/contribution here. Also should give statistics (like successful rate) of the results. Comments on failed cased and future work. Also should list all results in the appendices.

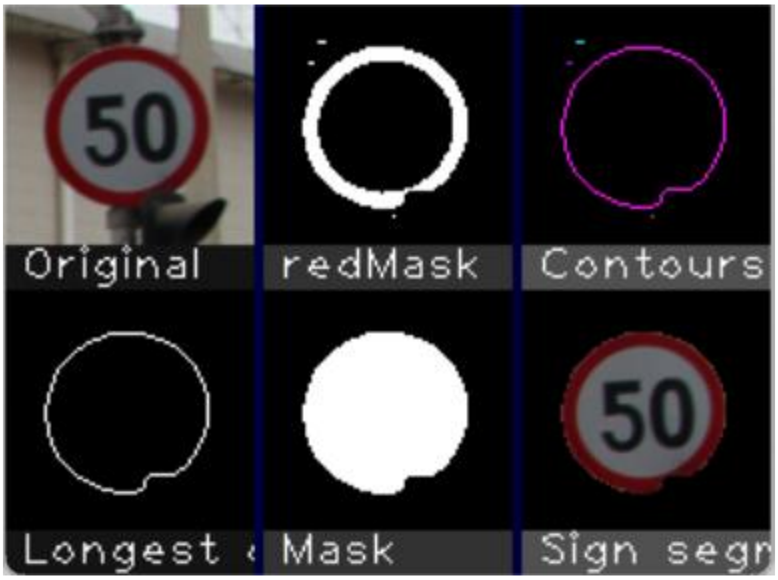

Figure 4.1.3. An image with an occluded traffic sign.

4.1.2 Results

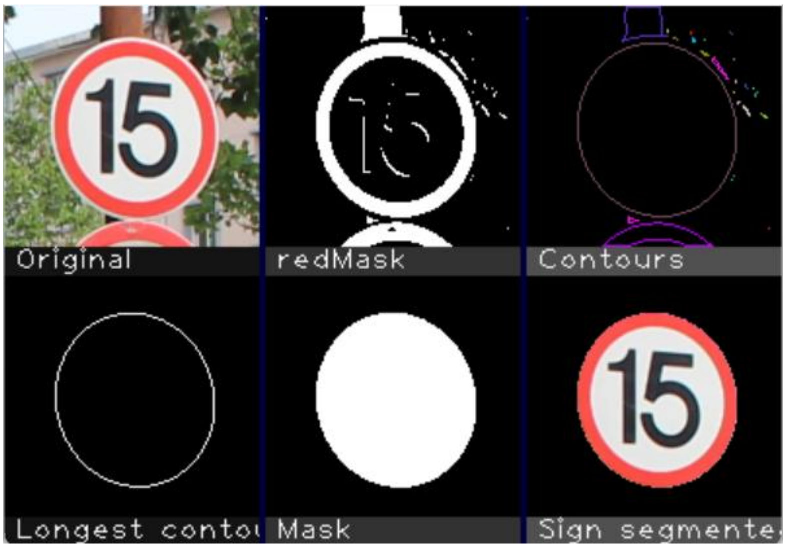

Figure 4.1.4. Segmentation results for 001_1_0009.png.

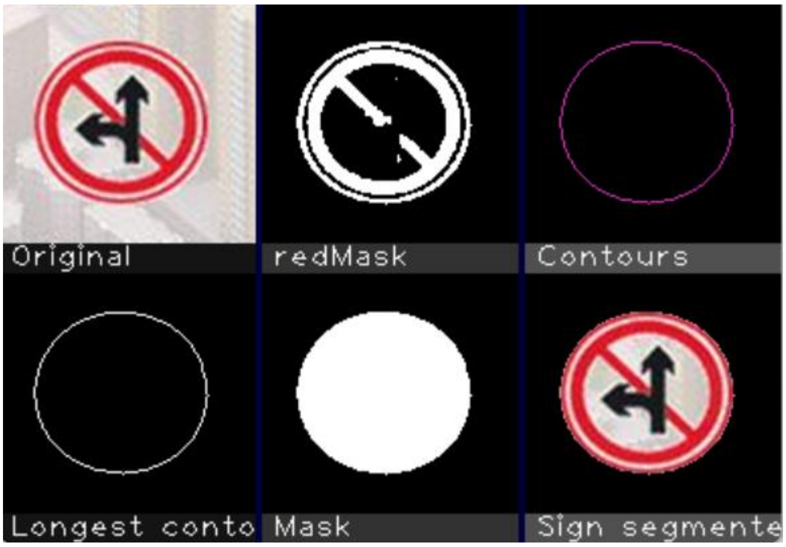

Figure 4.1.5. Segmentation results for 008_1_0008_1_j.png.

4.2 Coding work 2 & results (blue color) (Refer to Appendix 2)

Cannot find.

Done by: Ang Jia Pei

4.2.1 Coding work (Refer to Appendix 2)

Explanation

In the traffic sign detection project, it aimed to recognize the traffic sign according to the color and shape of the sign. Hence, a blue color processing algorithm had been implemented to perform segmentation on the blue-dominant regions within the blue traffic sign images. The images of blue traffic signs were imported from the blue sign file, and then the split() function was applied to split the images into their respective BGR (Blue, Green, Red) components.

A blue mask was created to highlight the regions dominated by blue color. Then,

Need detail.

a condition, which is that the intensity of the blue channel was significantly greater than both the red and green channels, was being set. The findContours() function was applied to discover the contours of the segmented blue regions; the contours with the longest boundary were assumed to represent the traffic sign.

For the detected traffic sign, the floodFill() function had been implemented to fill up the contour in order to ensure the isolation of the region within the boundary. Then, the segmented region was overlaid on the original image to extract the blue traffic sign while filtering off the remaining regions. The output windows were shown with the combination of original, blueMask, contours, longest contour, mask and sign segmented, presenting a clear picture of the processed result.

As a result, the project's preliminary implementation relied on assumptions such as the longest contour representing the traffic sign. Although the results showed that this method could segment the blue traffic signs successfully, it posed some limitations, such as potential inaccuracies due to the light condition, reflection, blurry images, damaged traffic signs, and colour fading on the traffic signs. To enhance the accuracy of traffic sign segmentation, it may need integration of shape information and fine-tuning the color segmentation process.

4.2.2 Result

<u>Segmentation results for blue color traffic signs:</u>

Figure 4.2.1 Segmentation results for 020_0003.png

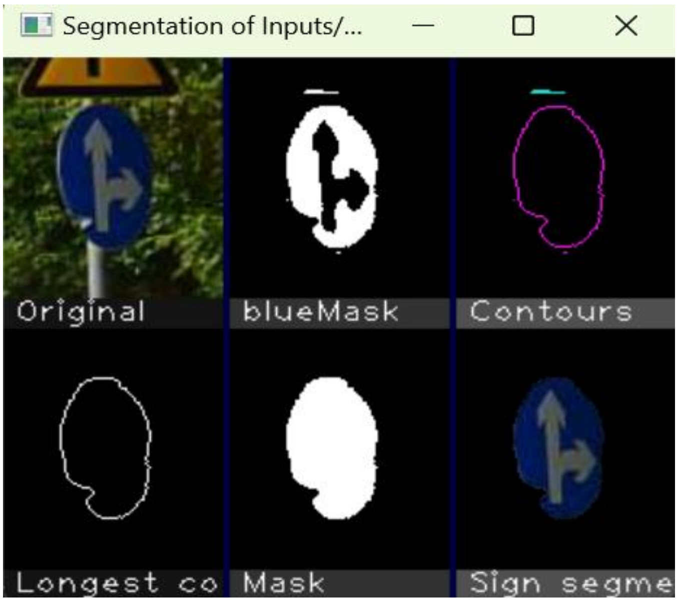

Figure 4.2.2 Segmentation results for partial damaged traffic sign, 020_0002_j.png

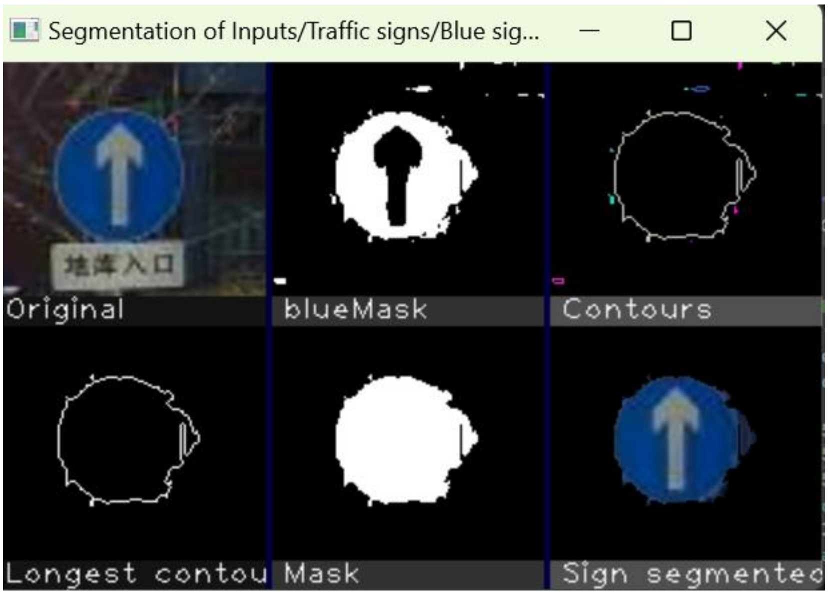

Figure 4.2.3 Segmentation results for blur traffic sign, 020_0002_j.png

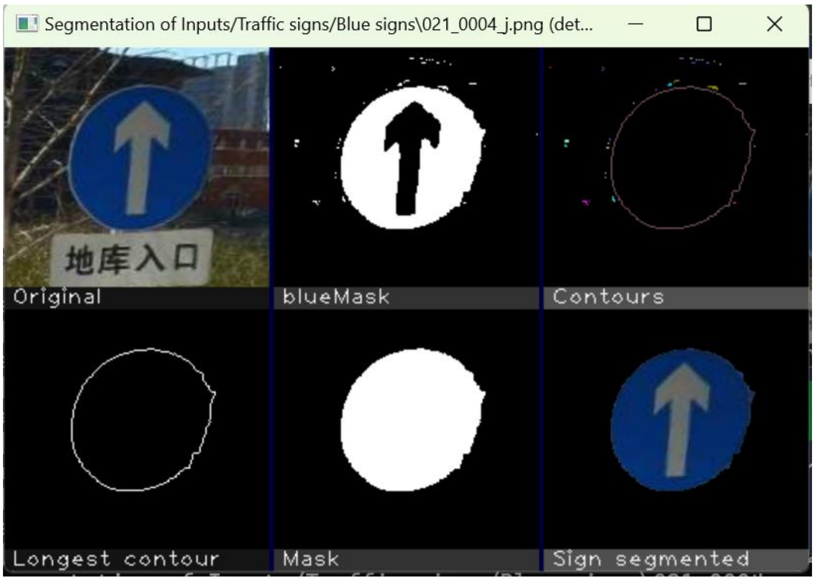

Figure 4.2.4 Segmentation results for 021_1_0004_1_j.png

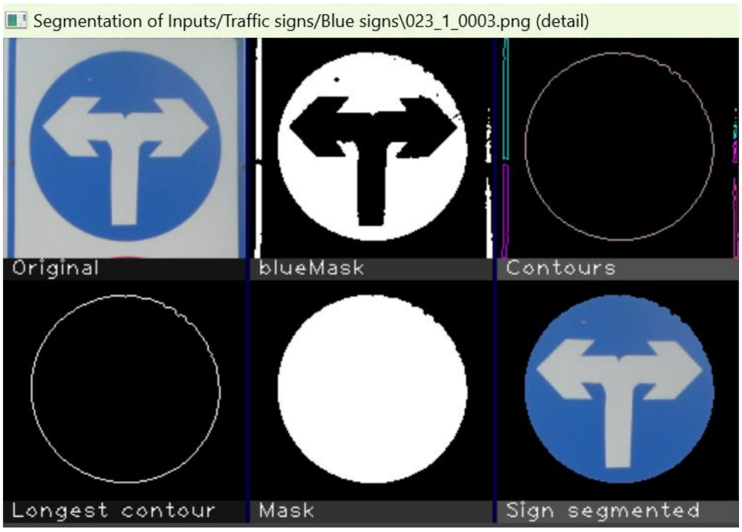

Figure 4.2.5 Segmentation results for 023_1_0003.png

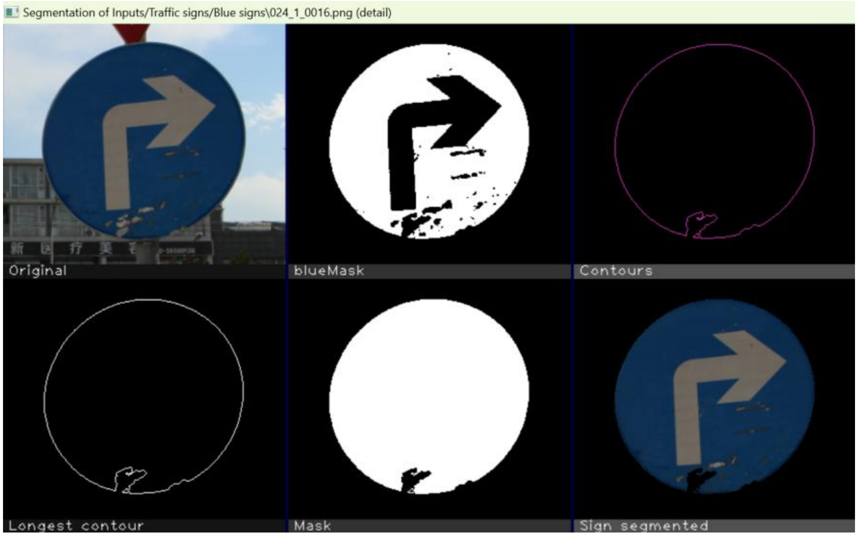

Figure 4.2.6 Segmentation results for 024_1_0016.png

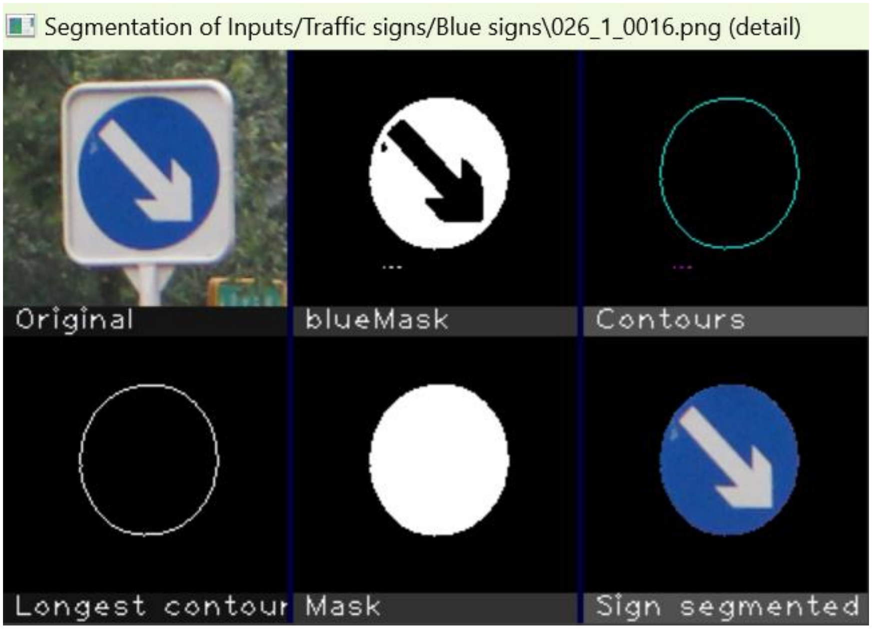

Figure 4.2.7 Segmentation results for 026_1_0016.png

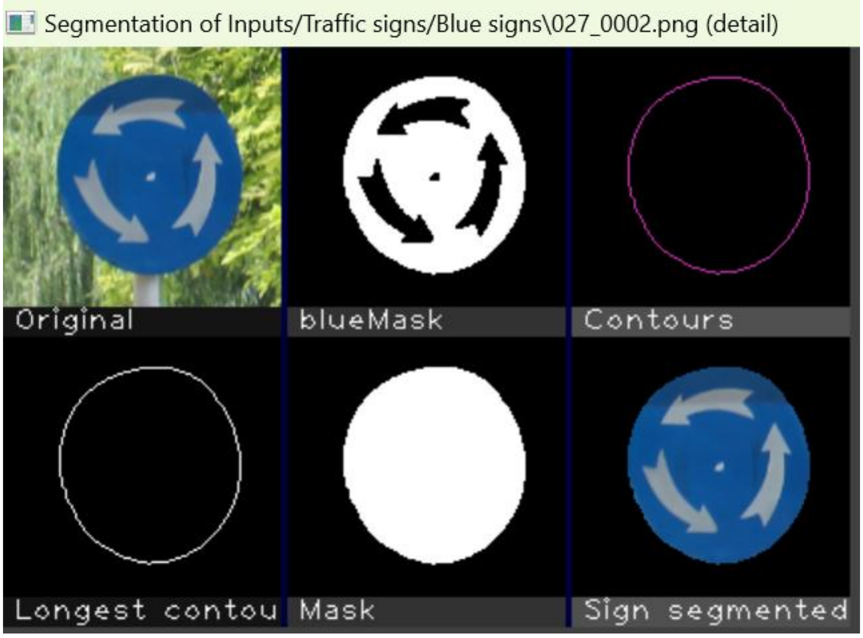

Figure 4.2.8 Segmentation results for 027_0002.png

Need more detail to emphasize your effort/contribution here. Also should give statistics (like successful rate) of the results. Comments on failed cased and future work. Also should list all results in the appendices.

4.3 Coding work 3 & results (yellow colour)

Done by Kok Yung Thow

No verb.

The preliminary work the segmentation of yellow traffic sign from images using HSV colour space. The lower boundary (Hue: 20, Saturation: 48.6%, Value: 11.7%) and upper boundary (Hue: 76, Saturation: 100%, Value:100%) were picked up from 2

Automate this process on some training data is desirable.

ground truth images with the lowest hue, saturation and value and the highest hue, saturation and value numbers of the yellow sign that were manually inspected in painting software, see Figure 4.3.1. The result was 86% segmentation rate (24 out of 28 images). Figure 4.3.2 – Figure 4.3.5 show the fail segmentation results with the corresponding original image, yellow mask, contour/edge, longest contour, mask and the sign segmented. The source code is attached in Appendix Coding work of 4.3.

Figure 4.3.1: Manually inspect the yellow sign colour of samples using Krita.

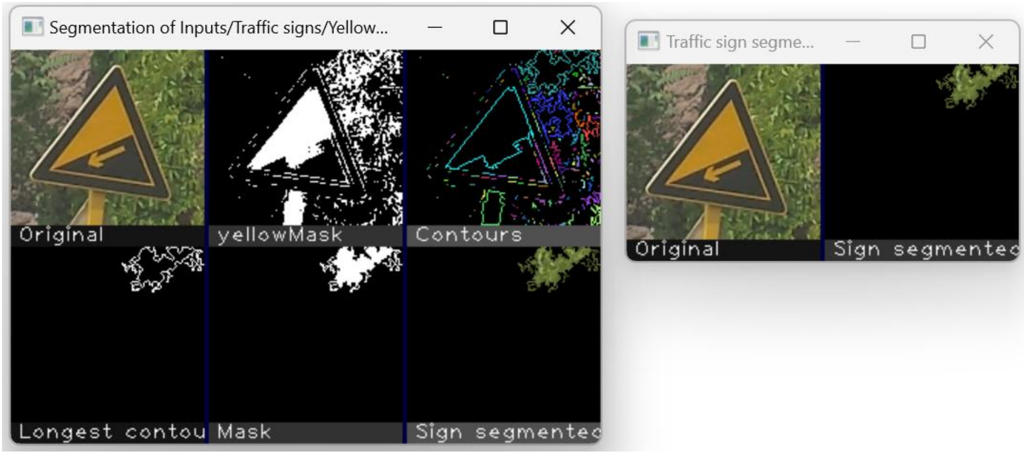

Figure 4.3.2: Fail segmentation result 1

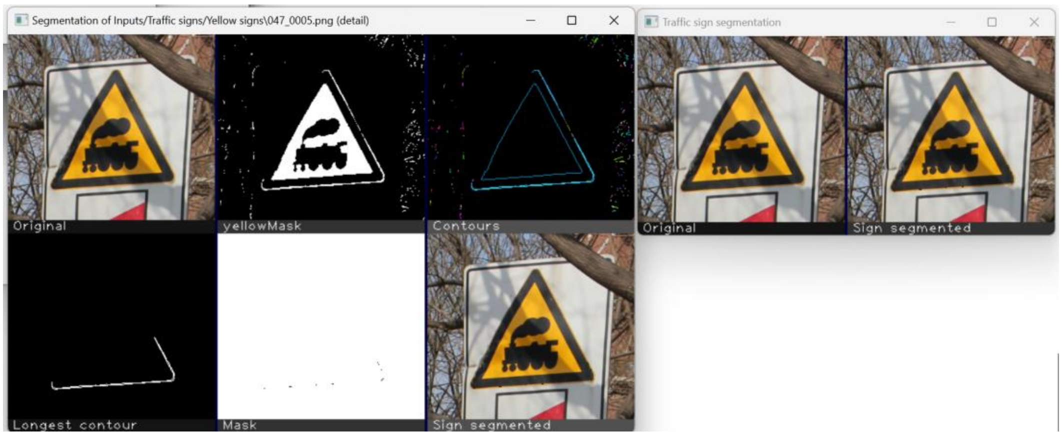

Figure 4.3.3: Fail Segmentation result 2

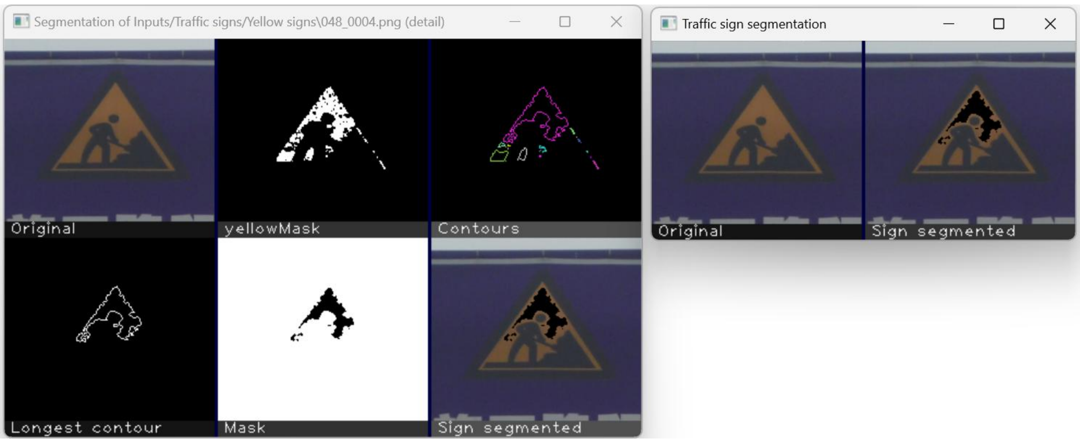

Figure 4.3.4: Fail Segmentation result 3

Figure 4.3.5: Fail segmentation result 4

The reason of these failures was caused by the lighting conditions that effected the saturations and values of the yellow colour. The possible solution for improvement is the HSI colour space conversion or integrate a triangle shape segmentation method. The succeed segmentation are shown in Figure 4.3.6 – 4.3.8.

Figure 4.3.6: Successed segmentation results 1

Need more comments on failed cased and future work.

Figure 4.3.7: Successed segmentation results 2

Figure 4.3.8: Successed segmentation results 3

4.4 Coding work 4 & results (shape)

Done by: Cheng Puei Ying

4.4.1 Coding work (Refer to Appendix 2)

Explanation

Better to have a flowchart for this part.

The program begins by **loading an image of a traffic sign**, which is stored in a matrix object. It then checks if the image is loaded correctly; if not, the program displays an error message and exits. Once the image is successfully loaded, it is **converted from the RGB color space to the HSV color space**. The HSV color space is particularly useful for image processing tasks because it separates the image intensity from color information, making it more robust to varying lighting conditions.

The text highlighted above is the un-important detail that makes your writing unnecessary long.

After converting the image to HSV, the program **splits the image into its three channels**: Hue, Saturation, and Value. The saturation channel is then processed using adaptive thresholding, which converts the image into a binary format where pixels are either black or white based on their intensity. This binary image undergoes **Gaussian blurring** to reduce noise and smooth the edges, which helps in better edge detection in subsequent steps.

The program then applies the **Canny edge detection algorithm** to identify the edges in the binary image. Once the edges are detected, the program uses the findContours function to identify the contours within the image. Contours represent the boundaries of objects within the image, and in this context, they correspond to the outlines of traffic signs.

See if can do a better summarizing job to cut short the description.

To determine the shape of the detected contours, the program **approximates each contour** to a polygon using the approxPolyDP function. This function simplifies the contour by reducing the number of vertices, making it easier to classify the shape. The program then **analyzes the number of vertices** in each polygon to classify the shape as a triangle, square, rectangle, circle, or octagon. For shapes with more than four vertices, the program checks the circularity of the contour to differentiate between circles and polygons.

The program then **identifies the largest relevant contour**, which is assumed to represent the traffic sign, and creates a mask for this contour. The mask is used to segment the traffic sign from the rest of the image, isolating it on a black background. The program then resizes all the images (the original image, contours, mask, and segmented sign) to a fixed size of 300x300 pixels.

Finally, the program **displays all the processed images** in a single window, divided into six sub-windows: the original image, contours, the longest contour, the mask, the detected shape with its name, and the segmented traffic sign. The program waits for a key press before closing the window and terminating.

Another example of verbose writing.

This coding successfully detected 19 traffic signs out of 26 images, resulting in an accuracy of approximately **73.08%**. To further increase the accuracy of detection, **integrating color detection** can be highly effective by leveraging color filtering to preprocess the images and isolate traffic signs from the background. By creating masks for these colors, the system can filter out irrelevant regions and focus on potential traffic signs.

4.4.2 Result

1. Circular road sign

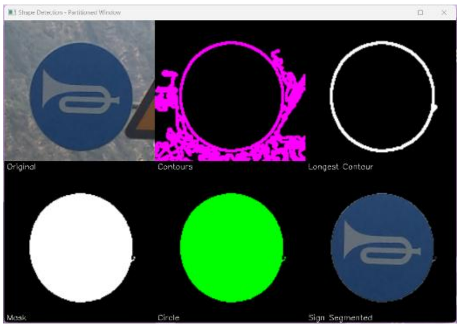

Figure 4.4.1 Segmentation result for Inputs/Traffic signs/Blue signs/029_1_0015.png

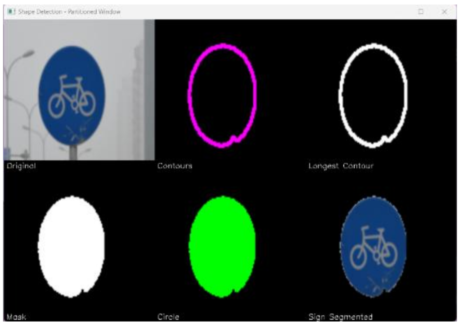

Figure 4.4.2 Segmentation result for Inputs/Traffic signs/Blue signs/030_1_0023.png

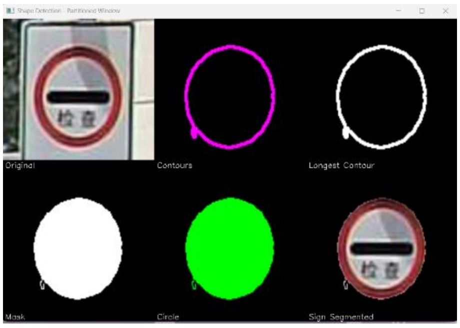

Figure 4.4.3 Segmentation result for Inputs/Traffic signs/Red Signs/057_0003_j.png

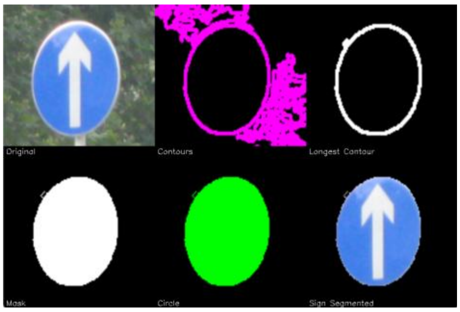

Figure 4.4.4 Segmentation result for Inputs/Traffic signs/Blue signs/021_0003.png

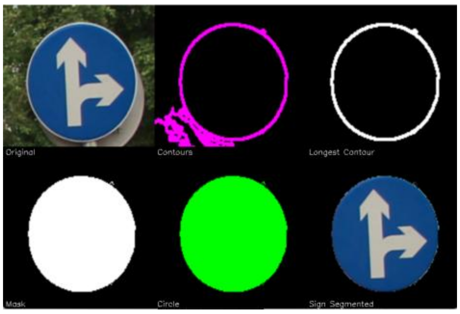

Figure 4.4.5 Segmentation result for Inputs/Traffic signs/Blue signs/020_0004.png

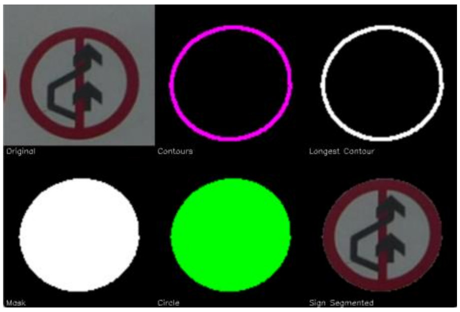

Figure 4.4.6 Segmentation result for Inputs/Traffic signs/Red Signs/014_1_0028.png

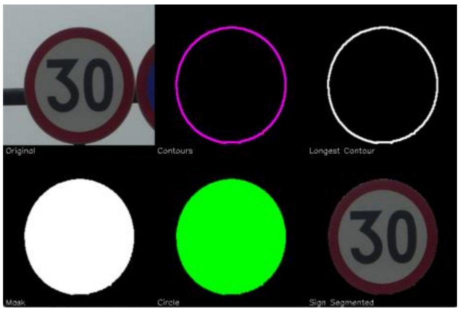

Figure 4.4.7 Segmentation result for Inputs/Traffic signs/Red Signs/002_0036.png

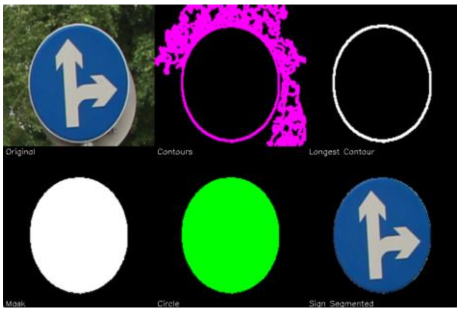

Figure 4.4.8 Segmentation result for Inputs/Traffic signs/Blue Signs/020_0003.png

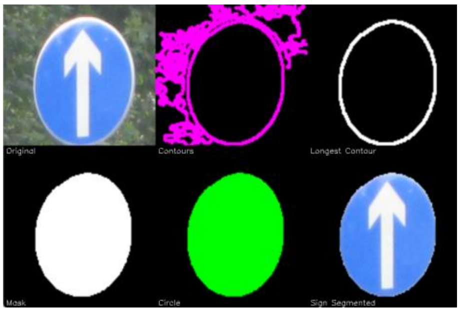

Figure 4.4.9 Segmentation result for Inputs/Traffic signs/Blue Signs/021_0004.png

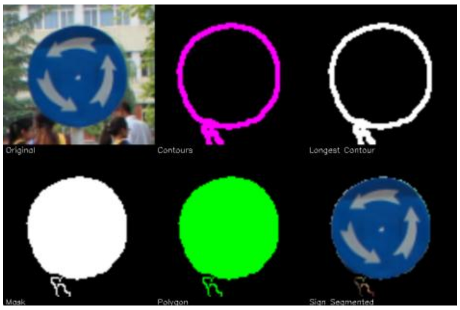

Figure 4.4.10 Segmentation result for Inputs/Traffic signs/Blue Signs/027_0010.png

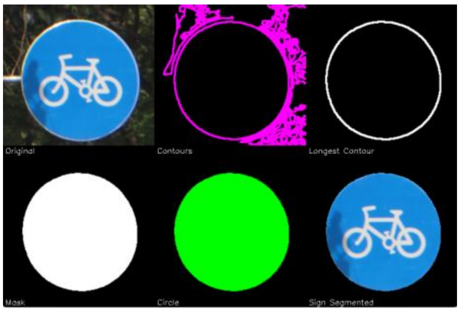

Figure 4.4.11 Segmentation result for Inputs/Traffic signs/Blue Signs/030_1_0002.png

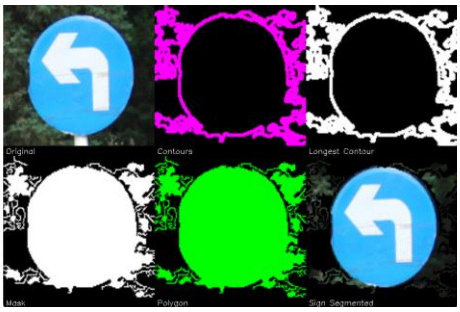

Figure 4.4.12 Segmentation result for Inputs/Traffic signs/Blue Signs/ 022_0005.png

2. Triangle road sign

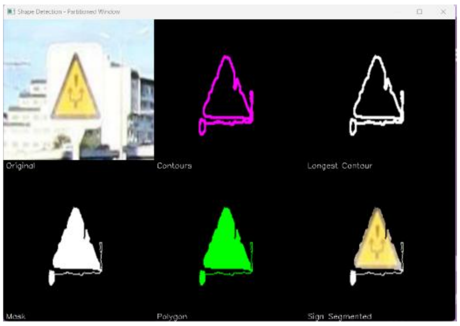

Figure 4.4.13 Segmentation result for Inputs/Traffic signs/Yellow Signs/032_1_0001_1_j.png

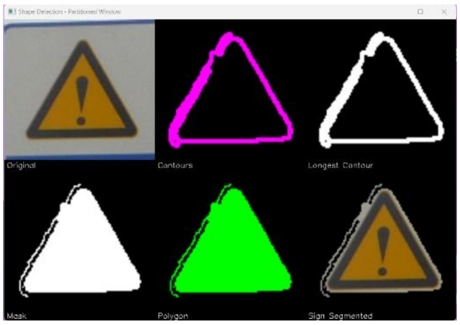

Figure 4.4.14 Segmentation result for Inputs/Traffic signs/Yellow Signs/034_1_0002.png

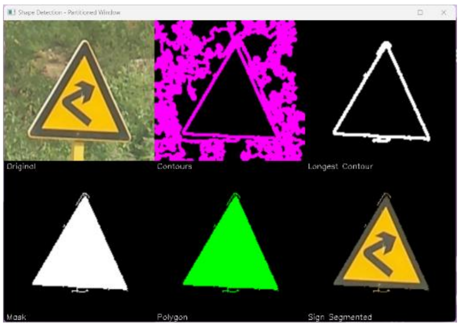

Figure 4.4.15 Segmentation result for Inputs/Traffic signs/Yellow Signs/038_0008.png

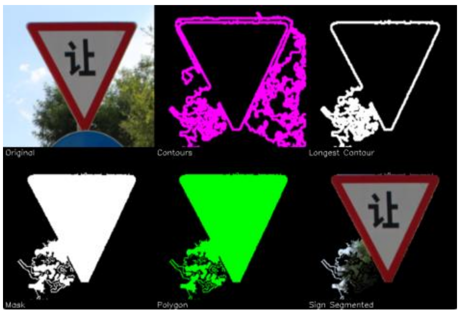

Figure 4.4.16 Segmentation result for Inputs/Traffic signs/Red Signs/056_1_0010.png

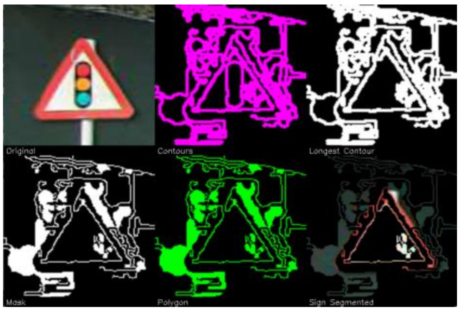

Figure 4.4.17 Segmentation result for Inputs/Traffic signs/Red Signs/033_1_0001.png

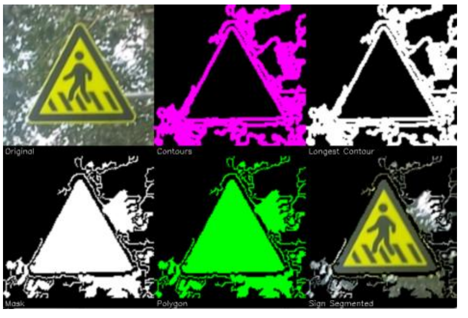

Figure 4.4.18 Segmentation result for Inputs/Traffic signs/Yellow Signs/035_1_0004.png

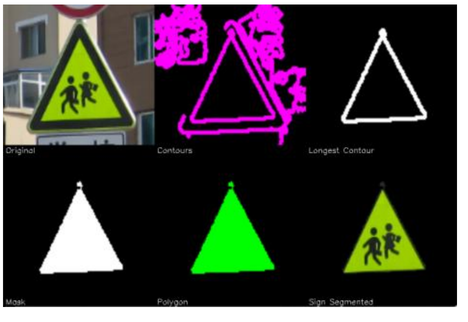

Figure 4.4.19 Segmentation result for Inputs/Traffic signs/Yellow Signs/037_0019.png

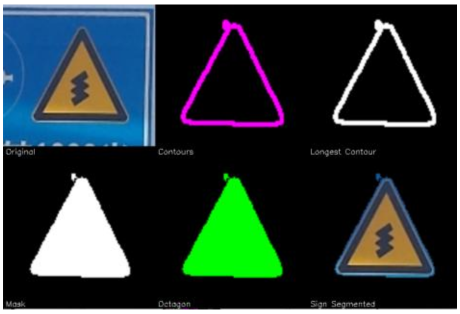

Figure 4.4.20 Segmentation result for Inputs/Traffic signs/Yellow Signs/049_1_0010.png

Figure 4.4.21 Segmentation result for Inputs/Traffic signs/Yellow Signs/035_1_0002.png

3. Rectangle road sign

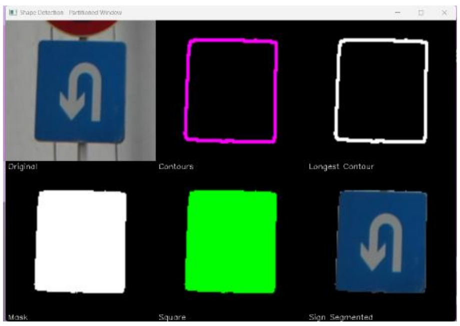

Figure 4.4.22 Segmentation result for Inputs/Traffic signs/Blue signs/031_0011.png

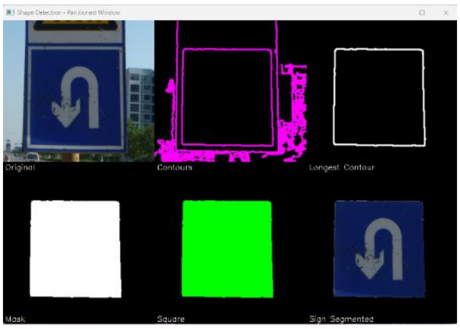

Figure 4.4.23 Segmentation result for Inputs/Traffic signs/Blue Signs/031_1_0020.png

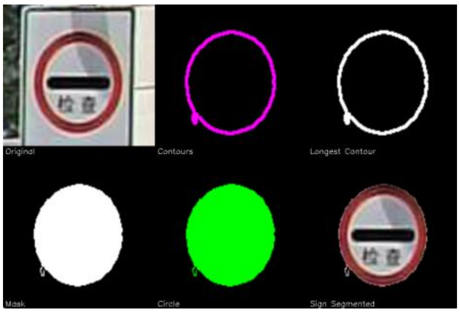

Figure 4.4.24 Segmentation result for Inputs/Traffic signs/Red Signs/057_0003_j.png

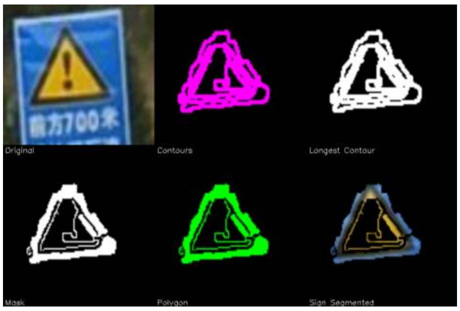

Figure 4.4.25 Segmentation result for Inputs/Traffic signs/Yellow Signs/034_1_0003_1_j.png

4. Octagon road sign

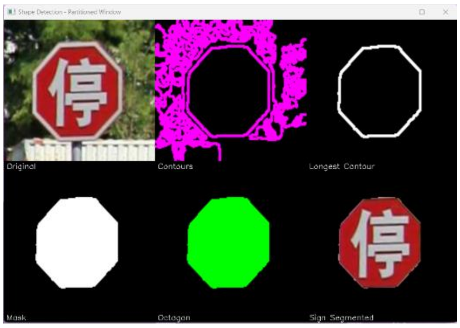

Figure 4.4.26 Segmentation result for Inputs/Traffic signs/Red Signs/052_1_0008_1_j.png

Need more detail to emphasize your effort/contribution here. Also should give statistics (like successful rate) of the results. Comments on failed cased and future work. Also should list all results in the appendices.

CHAPTER 5 - Conclusion

5.1 Summarization of Finding

Done by: Tan Pei Shi

In conclusion, traditional methods of traffic sign detection have struggled with complex environmental factors, such as occlusions, complex backgrounds, illumination changes, and adverse weather conditions. These factors have contributed to the issues of inaccurate or misdetection of signs. Hence, our project aims to develop robust traffic sign detection to handle these challenging factors using image processing and machine learning techniques. Accurate traffic sign detection can significantly improve the reliability of autonomous vehicles, thereby reducing the risk of accidents caused by inaccurate detection.

In order to ensure that the proposed system can accurately process traffic signs under various conditions, a hybrid approach has been proposed, whereby a template-based method is integrated with a multiscale attention mechanism. The template-based method provides the ability to predict the shape of occluded signs based on the information from similar signs. In contrast, the multiscale attention mechanism allows the system to focus on important features at multiple scales. The cooperation between the template-based method and multiscale attention mechanism can result in lower computational complexity as feature extraction can be performed directly on the ROI extracted by the template-based method. Thus, enhancing the accuracy of detection.

5.2 Novelties of work

Done by: Kok Yung Thow

The proposed TSDR system applied a hybrid approach of template matching and multiscale attention mechanism with low computational requirements and extremely high performance in detecting and recognizing traffic signs. Template matching can reduce the computational cost in generating the ROIs. The ROIs then will pass to multiscale attention mechanism to map and weight with the feature maps before the classification. This combination is an innovative and reliable approach in solving the real time TSDR system for autonomous vehicles that can detect and interpret the traffic signs in real-world with adverse conditions.

5.3 Concluding with Supportive Remark

Done by: Cheng Puei Ying

This project aimed to develop a reliable and efficient TSDR system, **leveraging color and shape-based segmentation techniques** for real-time operation in autonomous driving systems. The project successfully identified the challenges and limitations inherent in the segmentation and recognition processes of traffic signs, particularly under varying conditions such as lighting, occlusions, and color similarities with the background.

In the initial stages of the project, significant progress was made in segmenting traffic signs based on their dominant colors, red, blue, and yellow. The preliminary work demonstrated that while segmentation accuracy was high, certain limitations, such as misidentification due to **similar background colors** and **incomplete segmentation of occluded signs**, required further refinement. The proposed solutions, including the **integration of shape-based segmentation** and the use of **template-based methods**, showed promise in overcoming these challenges.

Besides, the implementation phase highlighted the importance of **high-quality equipment**, such as high-resolution cameras and external GPU units, to reduce training time and enhance the overall performance of the TSDR system. Moreover, the **involvement of volunteers** to provide diverse road scene data also played a crucial role in improving the robustness and accuracy of the system.

In conclusion, the project's findings underscore the feasibility of developing a real-time TSDR system that can be integrated into autonomous driving systems. However, the need for continued refinement, particularly in addressing the identified challenges, is essential for achieving optimal performance. The collaborative efforts, advanced hardware, and innovative approaches adopted in this project lay a strong foundation for future advancements in traffic sign detection and recognition technology.

This work has not only contributed valuable insights into the field of TSDR but also highlighted the critical role of continuous research and development in enhancing the capabilities of autonomous driving systems. Future work will focus on further optimizing the segmentation algorithms, exploring advanced machine learning techniques, and expanding the dataset to include more diverse and challenging traffic sign scenarios.

Turnitin report?

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